Extended Multiobject Tracking with Nonparametric Bayesian Learning of Object Classes and Model Parameters: Supplementary Material

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This manuscript provides detailed derivations as well as background and additional material for the publication "Extended Multiobject Tracking with Nonparametric Bayesian Learning of Object Classes and Model Parameters" [1], hereafter referred to as "main paper." The section, figure, and equation labels in this manuscript contain the prefix "S", e.g., (S.1). When sections and equations are referenced without this prefix, they can be found in the main paper. Notation and abbreviations established in the main paper will typically be used without redefining them.

S1 Statistical Independence Properties

We first list and, in some cases, briefly discuss the independence and conditional independence properties that are inherent in the system model presented in Sections II and III of the main paper. The independence of random vectors $\mathbf{a}$ and $\mathbf{b}$ is expressed by $\mathbf{a} \perp\!\!\!\perp \mathbf{b}$, and the conditional independence of $\mathbf{a}$ and $\mathbf{b}$ given $\mathbf{c}$ is expressed by $\mathbf{a} \perp\!\!\!\perp \mathbf{b} | \mathbf{c}$. The main independence structures are illustrated in the Bayesian networks [2] depicted in Fig. S1. These two Bayesian networks are from different viewpoints, and we will summarize in the following the (conditional) independence properties expressed by each network that are required in our derivations in Sections S2–S5. We start with the (conditional) independence properties between objects as expressed by Fig. S1(a).

A.1) $\mathbf{p}^*_c \perp\!\!\!\perp \mathbf{c}, \mathbf{P}^*_{\neg c}$, for any $c \in \mathbb{N}$, where $\mathbf{P}^*_{\neg c}$ is the matrix consisting of vectors $\mathbf{p}^*_{c'}$ for all $c' \in \mathcal{C} \setminus \{c\}$ with the set $\mathcal{C} \triangleq \{c_1, c_2, \ldots, c_I\}$. This is a core assumption of DP models [3,4].

A.2) $\mathbf{X}_{\mathcal{K}_i,i}, \mathbf{Z}_{\mathcal{K}_i,i} \perp\!\!\!\perp \mathbf{X}_{\mathcal{K}_{i'},i'}, \mathbf{Z}_{\mathcal{K}_{i'},i'} \,\big|\, \mathbf{p}^*_c, \mathbf{c}, \mathbf{P}^*_{\neg c}$, for $i' \neq i$.

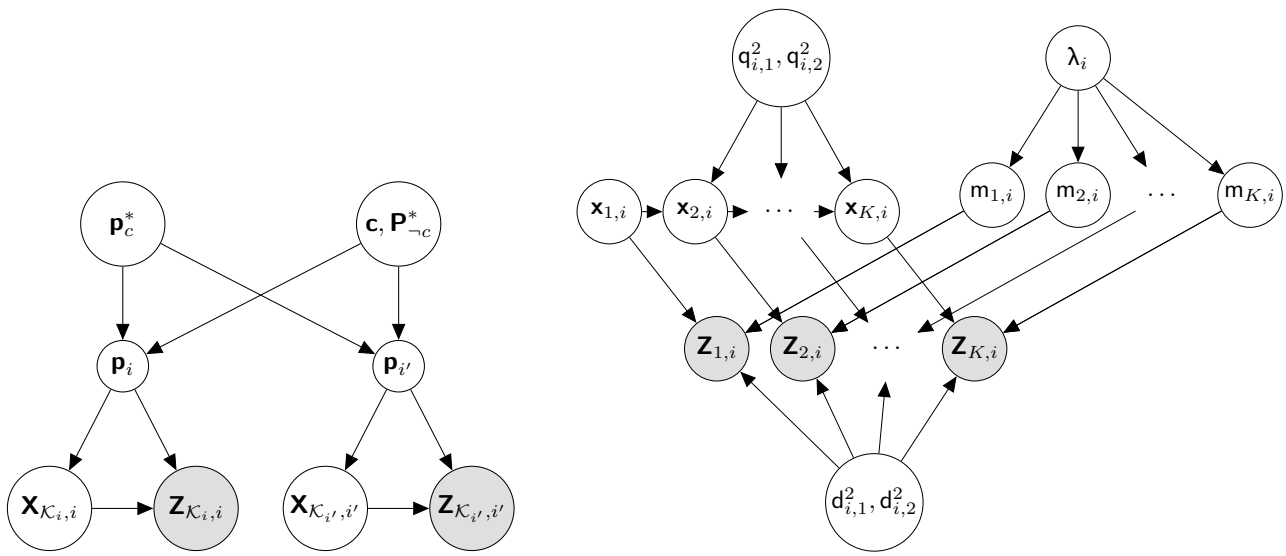
A.3) $\mathbf{X}_{\mathcal{K}_i,i}, \mathbf{Z}_{\mathcal{K}_i,i} \perp\!\!\!\perp \mathbf{p}^*_c, \mathbf{c}, \mathbf{P}^*_{\neg c} \,\big|\, \mathbf{p}_i$.

Next, we list the (conditional) independence properties of the random variables associated with a single object, as expressed by Fig. S1(b).

B.1) $\mathbf{X}_{\mathcal{K}_i,i} \perp\!\!\!\perp \mathsf{d}^2_{i,1}, \mathsf{d}^2_{i,2}, \lambda_i \,\big|\, \mathsf{q}^2_{i,1}, \mathsf{q}^2_{i,2}$.

B.2) $\mathbf{Z}_{\mathcal{K}_i,i} \perp\!\!\!\perp \mathsf{q}^2_{i,1}, \mathsf{q}^2_{i,2}, \lambda_i \,\big|\, \mathbf{X}_{\mathcal{K}_i,i}, \mathbf{m}_{\mathcal{K}_i,i}, \mathsf{d}^2_{i,1}, \mathsf{d}^2_{i,2}$, where $\mathbf{m}_{\mathcal{K}_i,i} \triangleq [\mathsf{m}_{k,i}]_{k \in \mathcal{K}_i}$.

B.3) $\mathbf{m}_{\mathcal{K}_i,i} \perp\!\!\!\perp \mathbf{X}_{\mathcal{K}_i,i}, \mathsf{q}^2_{i,1}, \mathsf{q}^2_{i,2}, \mathsf{d}^2_{i,1}, \mathsf{d}^2_{i,2} \,\big|\, \lambda_i$.



(a) (b)

Figure S1: Two Bayesian networks representing the (conditional) independence structure inherent in the system model presented in Sections II and III: (a) Structure across objects i and i' for $i' \neq i$, (b) structure of the marginal distribution of object i. In (a), c is any fixed class index from $\mathbb{N}$. In (b), for simplicity of presentation, we assume that $k_{\mathrm{S},i} = 1$ and $k_{\mathrm{E},i} = K$. Note that in some cases, conditioning on the parent nodes results in degenerate distributions (e.g., conditioned on $\mathbf{p}_c^*$, $\mathbf{P}_{\neg c}^*$, and $\mathbf{c}$, the random variable $\mathbf{p}_i$ is deterministically equal to $\boldsymbol{p}_{c_i}^*$).

B.4) $\mathbf{Z}_{k,i} \perp\!\!\!\perp \mathbf{Z}_{k',i}, \mathsf{m}_{k',i}, \mathbf{x}_{k',i} \mid \mathsf{m}_{k,i}, \mathbf{x}_{k,i}, \mathsf{d}_{i,1}^2, \mathsf{d}_{i,2}^2$, for $k' \neq k$.

B.5) $\mathbf{x}_{k,i} \perp\!\!\!\perp \mathbf{X}_{k_{\mathrm{S},i}:k-2,i}, \mathsf{d}_{i,1}^2, \mathsf{d}_{i,2}^2, \lambda_i \mid \mathbf{x}_{k-1,i}, \mathsf{q}_{i,1}^2, \mathsf{q}_{i,2}^2$, for $k \geq k_{\mathrm{S},i}+2$, where $\mathbf{X}_{k_1:k_2,i} \triangleq \left[\mathbf{x}_{k,i}\right]_{k=k_1}^{k_2}$. In particular, the conditional independence of $\mathbf{x}_{k,i}$ and $\mathbf{X}_{k_{\mathrm{S},i}:k-2,i}$ given $\mathbf{x}_{k-1,i}, \mathsf{q}_{i,1}^2$, and $\mathsf{q}_{i,2}^2$ is a common assumption in parameter-dependent state-space models [5, 6]. It means that the motion model described in Section II-A constitutes a Markov process, i.e., the state of an object is conditionally independent of all other past object states when given the object state at the previous time step, $\boldsymbol{x}_{k-1,i}$, and the object's driving noise covariance matrix $\boldsymbol{Q}_i = \mathrm{diag}(q_{i,1}^2 \mathbf{I}_2, q_{i,2}^2 \mathbf{I}_2)$. Here, we extend the assumption to include the conditional independence of $\mathbf{x}_{k,i}$ and $\mathsf{d}_{i,1}^2, \mathsf{d}_{i,2}^2, \lambda_i$ given $\mathbf{x}_{k-1,i}, \mathsf{q}_{i,1}^2$, and $\mathsf{q}_{i,2}^2$.

B.6) $\mathbf{x}_{k_{\mathrm{S},i},i} \perp\!\!\!\perp \mathbf{p}_i$, or, in detail, $\mathbf{x}_{k_{\mathrm{S},i},i} \perp\!\!\!\perp \mathsf{q}_{i,1}^2, \mathsf{q}_{i,2}^2, \mathsf{d}_{i,1}^2, \mathsf{d}_{i,2}^2, \lambda_i$.

B.7) $\mathbf{Z}_{k,i} \perp\!\!\!\perp \mathbf{Z}_{k_{\mathrm{S},i}:k-1,i}, \mathbf{X}_{k_{\mathrm{S},i}:k-1,i}, \mathbf{X}_{k+1:k_{\mathrm{E},i},i} \mid \mathbf{x}_{k,i}, \mathbf{p}_i$, where $\mathbf{Z}_{k_1:k_2,i} \triangleq \left[\mathbf{Z}_{k,i}\right]_{k=k_1}^{k_2}$. This is a common assumption made in the multiobject tracking literature using parameter-dependent state-space models [5, 6].

Besides these independence relations, we will also make use of the following prior distributions and likelihood functions.

P.1) The base distribution in our DP model, and thus also the prior for any $\boldsymbol{p}_i$ and $\boldsymbol{p}_c^*$, is given by (cf. (15) and (19))

$$f(\boldsymbol{p}_i) = f_{\mathrm{B}}(\boldsymbol{p}_i) = f(q_{i,1}^2, q_{i,2}^2) f(d_{i,1}^2, d_{i,2}^2) f(\lambda_i), \tag{S.1}$$

with (cf. Section III-C)

$$f(q_{i,1}^2, q_{i,2}^2) = \Gamma^{-1}\big(q_{i,1}^2; a_{\mathrm{q},1}, b_{\mathrm{q},1}\big)\, \Gamma^{-1}\big(q_{i,2}^2; a_{\mathrm{q},2}, b_{\mathrm{q},2}\big), \tag{S.2}$$

$$f(d_{i,1}^2, d_{i,2}^2) = \Gamma^{-1}\big(d_{i,1}^2; a_{\mathrm{d},1}, b_{\mathrm{d},1}\big)\, \Gamma^{-1}\big(d_{i,2}^2; a_{\mathrm{d},2}, b_{\mathrm{d},2}\big), \tag{S.3}$$

$$f(\lambda_i) = \Gamma\big(\lambda_i; a_\lambda, b_\lambda\big). \tag{S.4}$$

P.2) The initial state distribution is given as (cf. (1))

$$f(\boldsymbol{x}_{k_{\mathrm{S},i},i}) = \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \mathbf{0}_{4\times 1}, \boldsymbol{\Sigma}_{\mathrm{S}}). \tag{S.5}$$

P.3) The state evolution model is given as (cf. (4))

$$f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i}, q_{i,1}^2, q_{i,2}^2) = \mathcal{N}(\boldsymbol{x}_{k,i}; \boldsymbol{F}\boldsymbol{x}_{k-1,i}, \boldsymbol{Q}_i), \tag{S.6}$$

where we recall that $\boldsymbol{Q}_i = \mathrm{diag}(q_{i,1}^2\mathbf{I}_2, q_{i,2}^2\mathbf{I}_2)$.

P.4) Given λ_i, the numbers of measurements $\mathsf{m}_{k,i}$ are conditionally iid across $k \in \mathcal{K}_i$ and individually Poisson distributed (cf. Section II-B1 and (5)), i.e.,

$$p(\boldsymbol{m}_{\mathcal{K}_i,i}|\lambda_i) = \prod_{k\in\mathcal{K}_i} \mathrm{Pois}(m_{k,i}; \lambda_i). \tag{S.7}$$

P.5) The measurement model is given as (cf. (10))

$$f(\boldsymbol{Z}_{k,i}|m_{k,i}, \boldsymbol{x}_{k,i}, d_{i,1}^2, d_{i,2}^2) = \prod_{n=1}^{m_{k,i}} \mathcal{N}\big(\boldsymbol{z}_{k,i}^{(n)}; \boldsymbol{H}\boldsymbol{x}_{k,i}, \boldsymbol{R}(\boldsymbol{x}_{k,i})\boldsymbol{D}_i\boldsymbol{R}(\boldsymbol{x}_{k,i})^{\mathrm{T}}\big), \tag{S.8}$$

where $\boldsymbol{D}_i = \mathrm{diag}(d_{i,1}^2, d_{i,2}^2)$.

S2 Joint pdf $f\big(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{p}_i\big)$

As a preparation for deriving expressions of the conditional pdf $f(\boldsymbol{p}_c^*|\boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z})$ in Section S3 and of the conditional pmf $p_i(c; \ell)$ in Section S4, we will first derive some other pdf and pmf expressions. These expressions arise in the course of deriving an expression of the joint pdf $f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)$, which will itself be used in Section S4.2.

We can factor the joint pdf as

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{p}_i) = f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) f(\boldsymbol{p}_i). \tag{S.9}$$

The first factor in (S.9) can be expressed as

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = \sum_{\boldsymbol{m}\in\mathbb{N}_0^{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1}} f_{\mathbf{X}_{\mathcal{K}_i,i},\mathbf{Z}_{\mathcal{K}_i,i},\mathbf{m}_{\mathcal{K}_i,i}|\mathbf{p}_i}(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{m}|\boldsymbol{p}_i) \tag{S.10}$$

$$= f\big(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i} = [\mathrm{ncol}(\boldsymbol{Z}_{k,i})]_{k\in\mathcal{K}_i}|\boldsymbol{p}_i\big), \tag{S.11}$$

where we recall that $\mathbf{m}_{\mathcal{K}_i,i} = [\mathsf{m}_{k,i}]_{k\in\mathcal{K}_i}$ and $\mathrm{ncol}(\mathbf{Z}_{k,i})$ denotes the number of columns in the matrix $\mathbf{Z}_{k,i}$. We note that the sum in (S.10) collapses because $\mathbf{Z}_{\mathcal{K}_i,i} = \boldsymbol{Z}_{\mathcal{K}_i,i}$ implies $\mathbf{m}_{\mathcal{K}_i,i} = \boldsymbol{m}_{\mathcal{K}_i,i} = [\mathrm{ncol}(\boldsymbol{Z}_{k,i})]_{k\in\mathcal{K}_i}$ (cf. Section II-B4). From this point forward, $\boldsymbol{m}_{\mathcal{K}_i,i}$ is understood to be equal to $[\mathrm{ncol}(\boldsymbol{Z}_{k,i})]_{k\in\mathcal{K}_i}$. Thus, we write (S.11) as

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}|\boldsymbol{p}_i). \tag{S.12}$$

Using the chain rule, (S.12) can be further expressed as

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)\, p(\boldsymbol{m}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)\, f(\boldsymbol{X}_{\mathcal{K}_i,i}|\boldsymbol{p}_i). \tag{S.13}$$

Inserting (S.13) and (S.1) into (S.9) then yields

$$\begin{aligned} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{p}_i) &= f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, p(\boldsymbol{m}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \\ &\quad \times f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(q_{i,1}^2, q_{i,2}^2) f(d_{i,1}^2, d_{i,2}^2)\, f(\lambda_i) \\ &= ABC, \end{aligned} \tag{S.14}$$

with

$$\begin{aligned} A &\triangleq f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(q_{i,1}^2, q_{i,2}^2), \\ B &\triangleq f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(d_{i,1}^2, d_{i,2}^2), \\ C &\triangleq p(\boldsymbol{m}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(\lambda_i). \end{aligned}$$

In what follows, we derive expressions of the factors A, B, and C.

S2.1 Factor $A = f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(q_{i,1}^2, q_{i,2}^2)$

Using B.1, the factor A can be simplified to

$$A = f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2)\, f(q_{i,1}^2, q_{i,2}^2). \tag{S.15}$$

Using the chain rule, the first factor in (S.15) is given as

$$\begin{aligned} f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2) &= f(\boldsymbol{x}_{k_{\mathrm{S},i},i}|q_{i,1}^2, q_{i,2}^2) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} f(\boldsymbol{x}_{k,i}|\boldsymbol{X}_{k_{\mathrm{S},i}:k-1,i}, q_{i,1}^2, q_{i,2}^2) \\ &= f(\boldsymbol{x}_{k_{\mathrm{S},i},i}) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i}, q_{i,1}^2, q_{i,2}^2), \end{aligned} \tag{S.16}$$

where B.6 and B.5 were used in the last step. Inserting (S.5) and (S.6) into (S.16), we obtain

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2) = \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \mathcal{N}(\boldsymbol{x}_{k,i}; \boldsymbol{F}\boldsymbol{x}_{k-1,i}, \boldsymbol{Q}_i). \tag{S.17}$$

Substituting $\boldsymbol{u}_{k,i} = \boldsymbol{x}_{k,i} - \boldsymbol{F}\boldsymbol{x}_{k-1,i}$ (cf. (2)) and noting that $\mathcal{N}(\boldsymbol{x}_{k,i}; \boldsymbol{F}\boldsymbol{x}_{k-1,i}, \boldsymbol{Q}_i) = \mathcal{N}(\boldsymbol{u}_{k,i}; \boldsymbol{0}_{4\times1}, \boldsymbol{Q}_i)$, and recalling that $\boldsymbol{Q}_i = \mathrm{diag}(q_{i,1}^2\mathbf{I}_2, q_{i,2}^2\mathbf{I}_2)$, we further obtain

$$\begin{aligned} f(\boldsymbol{X}_{\mathcal{K}_i,i}|q_{i,1}^2, q_{i,2}^2) &= \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \mathcal{N}(\boldsymbol{u}_{k,i}; \boldsymbol{0}_{4\times1}, \boldsymbol{Q}_i) \\ &= \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \prod_{\kappa=1}^{2} \mathcal{N}(u_{k,i,\kappa}; 0, q_{i,1}^2)\, \mathcal{N}(u_{k,i,\kappa+2}; 0, q_{i,2}^2), \end{aligned} \tag{S.18}$$

where $u_{k,i,1}$ through $u_{k,i,4}$ are the elements of $\boldsymbol{u}_{k,i} = \boldsymbol{x}_{k,i} - \boldsymbol{F}\boldsymbol{x}_{k-1,i}$. Inserting (S.18) and (S.2) into (S.15) then gives

$$\begin{aligned} A &= \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}) \left(\prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \prod_{\kappa=1}^{2} \mathcal{N}(u_{k,i,\kappa}; 0, q_{i,1}^2)\, \mathcal{N}(u_{k,i,\kappa+2}; 0, q_{i,2}^2) \right) \\ &\quad \times \Gamma^{-1}(q_{i,1}^2; a_{\mathrm{q},1}, b_{\mathrm{q},1})\, \Gamma^{-1}(q_{i,2}^2; a_{\mathrm{q},2}, b_{\mathrm{q},2}). \end{aligned} \tag{S.19}$$

Because the inverse gamma distribution is a conjugate prior for a Gaussian likelihood function with unknown variance [7, Section 2.6], this is proportional to the product of two inverse gamma distributions. However, for our subsequent development (see Section S4.2), we require equality and not merely proportionality.

S2.2 Factor $B = f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(d_{i,1}^2, d_{i,2}^2)$

For the factor B, we obtain

$$B = f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2)\, f(d_{i,1}^2, d_{i,2}^2), \tag{S.20}$$

where the removal of $q_{i,1}^2$, $q_{i,2}^2$, and λ_i from the condition set of $f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)$ follows from B.2. Using the chain rule, the first factor in (S.20) becomes

$$\begin{aligned} f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) &= f(\boldsymbol{Z}_{k_{\mathrm{S},i},i}|\,\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) \\ &\quad \times \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} f(\boldsymbol{Z}_{k,i}|\,\boldsymbol{Z}_{k_{\mathrm{S},i}:k-1,i}, \boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2). \end{aligned} \tag{S.21}$$

Using B.4, we have $f(\boldsymbol{Z}_{k_{\mathrm{S},i},i}|\,\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) = f(\boldsymbol{Z}_{k_{\mathrm{S},i},i}|\boldsymbol{x}_{k_{\mathrm{S},i},i}, m_{k_{\mathrm{S},i},i}, d_{i,1}^2, d_{i,2}^2)$ and $f(\boldsymbol{Z}_{k,i}|\,\boldsymbol{Z}_{k_{\mathrm{S},i}:k-1,i}, \boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) = f(\boldsymbol{Z}_{k,i}|\boldsymbol{x}_{k,i}, m_{k,i}, d_{i,1}^2, d_{i,2}^2)$, and thus expression (S.21) simplifies to

$$f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) = \prod_{k\in\mathcal{K}_i} f(\boldsymbol{Z}_{k,i}|\boldsymbol{x}_{k,i}, m_{k,i}, d_{i,1}^2, d_{i,2}^2). \tag{S.22}$$

Inserting (S.8) into (S.22), we obtain

$$f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) = \prod_{k\in\mathcal{K}_i} \prod_{n=1}^{m_{k,i}} \mathcal{N}\big(\boldsymbol{z}_{k,i}^{(n)}; \boldsymbol{H}\boldsymbol{x}_{k,i}, \boldsymbol{R}(\boldsymbol{x}_{k,i})\boldsymbol{D}_i\boldsymbol{R}(\boldsymbol{x}_{k,i})^{\mathrm{T}}\big).$$

Substituting $\boldsymbol{v}_{k,i}^{(n)} = \boldsymbol{R}(\boldsymbol{x}_{k,i})^{-1}\big(\boldsymbol{z}_{k,i}^{(n)} - \boldsymbol{H}\boldsymbol{x}_{k,i}\big)$ (cf. (6)) and noting that $\mathcal{N}\big(\boldsymbol{z}_{k,i}^{(n)}; \boldsymbol{H}\boldsymbol{x}_{k,i}, \boldsymbol{R}(\boldsymbol{x}_{k,i})\boldsymbol{D}_i\boldsymbol{R}(\boldsymbol{x}_{k,i})^{\mathrm{T}}\big) = \mathcal{N}\big(\boldsymbol{v}_{k,i}^{(n)}; \boldsymbol{0}_{2\times1}, \boldsymbol{D}_i\big)$, and recalling that $\boldsymbol{D}_i = \mathrm{diag}(d_{i,1}^2, d_{i,2}^2)$, we further obtain

$$f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2) = \prod_{k\in\mathcal{K}_i} \prod_{n=1}^{m_{k,i}} \mathcal{N}\big(\boldsymbol{v}_{k,i}^{(n)}; \boldsymbol{0}_{2\times1}, \boldsymbol{D}_i\big) = \prod_{k\in\mathcal{K}_i} \prod_{n=1}^{m_{k,i}} \prod_{\kappa=1}^{2} \mathcal{N}\big(v_{k,i,\kappa}^{(n)}; 0, d_{i,\kappa}^2\big), \tag{S.23}$$

where $v_{k,i,1}^{(n)}$ and $v_{k,i,2}^{(n)}$ are the elements of $\boldsymbol{v}_{k,i}^{(n)}$. Finally, inserting (S.23) and (S.3) into (S.20) yields

$$B = \left(\prod_{k\in\mathcal{K}_i} \prod_{n=1}^{m_{k,i}} \prod_{\kappa=1}^{2} \mathcal{N}\big(v_{k,i,\kappa}^{(n)}; 0, d_{i,\kappa}^2\big)\right) \Gamma^{-1}(d_{i,1}^2; a_{\mathrm{d},1}, b_{\mathrm{d},1})\, \Gamma^{-1}(d_{i,2}^2; a_{\mathrm{d},2}, b_{\mathrm{d},2}). \tag{S.24}$$

Again, the proportionality to a product of two inverse gamma distributions is not useful at this point (see Section S4.2).

S2.3 Factor $C = p(\boldsymbol{m}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, f(\lambda_i)$

Using B.3, the factor C simplifies to

$$C = p(\boldsymbol{m}_{\mathcal{K}_i,i}|\lambda_i)\, f(\lambda_i).$$

By (S.7) and (S.4), we further obtain

$$C = \left(\prod_{k \in \mathcal{K}_i} \text{Pois}(m_{k,i}\,;\, \lambda_i) \right) \Gamma\big(\lambda_i\,; a_\lambda, b_\lambda\big). \tag{S.25}$$

At this point, we could use the fact that the gamma distribution is a conjugate prior for a Poisson likelihood function with unknown rate parameter λ_i [7, Section 2.6]. However, this is not helpful for our subsequent development (see Section S4.2).

S3 Conditional pdf $f(\boldsymbol{p}_c^* \,|\, \boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z})$

Next, we will derive expressions (27)–(31) for the conditional pdf $f(\boldsymbol{p}_c^* \,|\, \boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z})$ involved in the Gibbs sampler (see Sections IV-C2 and V-B). Throughout this section, we will use the symbol $\propto$ to designate proportionality up to a constant independent of $\boldsymbol{p}_c^* = \big[q_{c,1}^{*2}\ q_{c,2}^{*2}\ d_{c,1}^{*2}\ d_{c,2}^{*2}\ \lambda_c^*\big]^{\mathrm{T}}$.

Using Bayes' theorem, we have

$$f(\boldsymbol{p}_c^* \,|\, \boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z}) \propto f(\boldsymbol{X}, \boldsymbol{Z} | \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*)\, f(\boldsymbol{p}_c^* | \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*). \tag{S.26}$$

Using A.2, the first factor in (S.26) factorizes according to

$$f(\boldsymbol{X}, \boldsymbol{Z} | \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) = \prod_{i=1}^{I} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*). \tag{S.27}$$

Since $\mathbf{p}_i$ is deterministically fixed as $\boldsymbol{p}_i = \boldsymbol{p}_{c_i}^*$ given $\boldsymbol{p}_c^*$, $\boldsymbol{c}$, and $\boldsymbol{P}_{\neg c}^*$, we can add $\boldsymbol{p}_i$ as a further condition in each factor on the right-hand side, i.e.,

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) = f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{p}_i = \boldsymbol{p}_{c_i}^*, \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) = f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{p}_i = \boldsymbol{p}_{c_i}^*), \tag{S.28}$$

where we used A.3 in the last equality. Inserting (S.28) into (S.27), we obtain further

$$f(\boldsymbol{X}, \boldsymbol{Z} | \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) = \prod_{i=1}^{I} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{p}_i = \boldsymbol{p}_{c_i}^*) \propto \prod_{i\,:\,c_i = c} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{p}_i = \boldsymbol{p}_c^*), \tag{S.29}$$

where we used the fact that the product of all factors with $c_i \neq c$ is a constant independent of $\boldsymbol{p}_c^*$ (recall that our goal is to find $f(\boldsymbol{p}_c^* \,|\, \boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z})$, i.e., a pdf in $\boldsymbol{p}_c^*$). In view of expression (S.29), from this point forward, we only consider i such that $c_i = c$ and, hence, $\boldsymbol{p}_i$ is always considered to be $\boldsymbol{p}_c^*$. Inserting (S.13) into (S.29), we obtain further

$$f(\boldsymbol{X}, \boldsymbol{Z} | \boldsymbol{p}_c^*, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) \propto \prod_{i\,:\,c_i = c} f(\boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)\, p(\boldsymbol{m}_{\mathcal{K}_i,i} | \boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)\, f(\boldsymbol{X}_{\mathcal{K}_i,i} | \boldsymbol{p}_i). \tag{S.30}$$

Next, using A.1, the second factor in (S.26) simplifies as

$$f(\boldsymbol{p}_c^* | \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) = f(\boldsymbol{p}_c^*).$$

Using P.1, we obtain further

$$f(\boldsymbol{p}_c^* | \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*) = f(q_{c,1}^{*2}, q_{c,2}^{*2})\, f(d_{c,1}^{*2}, d_{c,2}^{*2})\, f(\lambda_c^*). \tag{S.31}$$

Finally, inserting (S.30) and (S.31) into (S.26) yields

$$
\begin{aligned}
&f(\boldsymbol{p}_c^* | \boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z}) \\
&\propto \Bigg(\prod_{i\,:\,c_i=c} f(\boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\, p(\boldsymbol{m}_{\mathcal{K}_i,i} | \boldsymbol{X}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \\
&\quad \times f(\boldsymbol{X}_{\mathcal{K}_i,i} | q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \Bigg) f(q_{c,1}^{*2}, q_{c,2}^{*2})\, f(d_{c,1}^{*2}, d_{c,2}^{*2})\, f(\lambda_c^*) \\
&= A' B' C',
\end{aligned}
\tag{S.32}
$$

with

$$
\begin{aligned}
A' &\triangleq \Bigg(\prod_{i\,:\,c_i=c} f(\boldsymbol{X}_{\mathcal{K}_i,i} | q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \Bigg) f(q_{c,1}^{*2}, q_{c,2}^{*2}), \\
B' &\triangleq \Bigg(\prod_{i\,:\,c_i=c} f(\boldsymbol{Z}_{\mathcal{K}_i,i} | \boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \Bigg) f(d_{c,1}^{*2}, d_{c,2}^{*2}), \\
C' &\triangleq \Bigg(\prod_{i\,:\,c_i=c} p(\boldsymbol{m}_{\mathcal{K}_i,i} | \boldsymbol{X}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \Bigg) f(\lambda_c^*).
\end{aligned}
$$

In these expressions, as mentioned previously, $q_{i,1}^2 = q_{c,1}^{*2}$ etc. In what follows, we derive expressions of the factors A', B', and C'.

S3.1 Factor $A' = \big(\prod_{i\,:\,c_i=c} f(\boldsymbol{X}_{\mathcal{K}_i,i} | q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i) \big) f(q_{c,1}^{*2}, q_{c,2}^{*2})$

Using B.1, the factor A' can be simplified to

$$
A' = \Bigg(\prod_{i\,:\,c_i=c} f(\boldsymbol{X}_{\mathcal{K}_i,i} | q_{i,1}^2, q_{i,2}^2) \Bigg) f(q_{c,1}^{*2}, q_{c,2}^{*2}).
$$

Inserting (S.18), using P.1, and recalling that $q_{i,1}^2 = q_{c,1}^{*2}$ and $q_{i,2}^2 = q_{c,2}^{*2}$, we obtain

$$
A' \propto \Bigg(\prod_{i\,:\,c_i=c} \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \prod_{\kappa=1}^{2} \mathcal{N}(u_{k,i,\kappa}; 0, q_{c,1}^{*2})\, \mathcal{N}(u_{k,i,\kappa+2}; 0, q_{c,2}^{*2}) \Bigg) \Gamma^{-1}(q_{c,1}^{*2}; a_{\mathrm{q},1}, b_{\mathrm{q},1})\, \Gamma^{-1}(q_{c,2}^{*2}; a_{\mathrm{q},2}, b_{\mathrm{q},2}).
\tag{S.33}
$$

The product of Gaussian distributions in (S.33) can be interpreted as a likelihood function for the unknown variances $\mathsf{q}_{c,1}^{*2}$ and $\mathsf{q}_{c,2}^{*2}$. Using the fact that the inverse gamma distribution is the conjugate prior for such a likelihood function [7, Section 2.6], we obtain the closed-form expression

$$
A' \propto \Gamma^{-1}(q_{c,1}^{*2}; a'_{\mathrm{q},c,1}, b'_{\mathrm{q},c,1})\, \Gamma^{-1}(q_{c,2}^{*2}; a'_{\mathrm{q},c,2}, b'_{\mathrm{q},c,2}),
\tag{S.34}
$$

with parameters

$$
a'_{\mathrm{q},c,\kappa} = a_{\mathrm{q},\kappa} + K(c) - N(c), \quad b'_{\mathrm{q},c,\kappa} = b_{\mathrm{q},\kappa} + \sum_{i\,:\,c_i=c} \sum_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \overline{u_{k,i,\kappa}^2}, \quad \kappa = 1, 2,
\tag{S.35}
$$

where $K(c)$ is the number of time steps k in which at least one existing object belongs to class c, $N(c)$ is the number of c_i that are equal to c, $\overline{u_{k,i,1}^2} \triangleq (u_{k,i,1}^2 + u_{k,i,2}^2)/2$, and $\overline{u_{k,i,2}^2} \triangleq (u_{k,i,3}^2 + u_{k,i,4}^2)/2$.

S3.2 Factor $B' = \big(\prod_{i\,:\,c_i=c} f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\big)\, f(d_{c,1}^{*2}, d_{c,2}^{*2})$

Using B.2, the factor B' can be simplified to

$$B' = \left(\prod_{i\,:\,c_i=c} f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{m}_{\mathcal{K}_i,i}, d_{i,1}^2, d_{i,2}^2)\right) f(d_{c,1}^{*2}, d_{c,2}^{*2}).$$

Inserting (S.23), using P.1, and recalling that $d_{i,1}^2 = d_{c,1}^{*2}$ and $d_{i,2}^2 = d_{c,2}^{*2}$, we obtain

$$B' = \left(\prod_{i\,:\,c_i=c} \prod_{k\in\mathcal{K}_i} \prod_{n=1}^{m_{k,i}} \prod_{\kappa=1}^{2} \mathcal{N}\big(v_{k,i,\kappa}^{(n)}; 0, d_{c,\kappa}^{*2}\big)\right) \Gamma^{-1}(d_{c,1}^{*2}; a_{\mathrm{d},1}, b_{\mathrm{d},1})\, \Gamma^{-1}(d_{c,2}^{*2}; a_{\mathrm{d},2}, b_{\mathrm{d},2}).$$

Again using the fact that the inverse gamma distribution is a conjugate prior for a Gaussian likelihood function with unknown variance [7, Section 2.6], we obtain

$$B' \propto \Gamma^{-1}(d_{c,1}^{*2}; a'_{\mathrm{d},c,1}, b'_{\mathrm{d},c,1})\, \Gamma^{-1}(d_{c,2}^{*2}; a'_{\mathrm{d},c,2}, b'_{\mathrm{d},c,2}), \tag{S.36}$$

with parameters

$$a'_{\mathrm{d},c,\kappa} = a_{\mathrm{d},\kappa} + \frac{1}{2}\sum_{i\,:\,c_i=c} M^{(i)}, \quad b'_{\mathrm{d},c,\kappa} = b_{\mathrm{d},\kappa} + \frac{1}{2}\sum_{i\,:\,c_i=c}\sum_{k\in\mathcal{K}_i}\sum_{n=1}^{m_{k,i}} (v_{k,i,\kappa}^{(n)})^2, \quad \kappa = 1, 2, \tag{S.37}$$

where

$$M^{(i)} \triangleq \sum_{k\in\mathcal{K}_i} m_{k,i}.$$

S3.3 Factor $C' = \big(\prod_{i\,:\,c_i=c} p(\boldsymbol{m}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, q_{i,1}^2, q_{i,2}^2, d_{i,1}^2, d_{i,2}^2, \lambda_i)\big)\, f(\lambda_c^*)$

Using B.3, the factor C' simplifies to

$$C' = \left(\prod_{i\,:\,c_i=c} p(\boldsymbol{m}_{\mathcal{K}_i,i}|\lambda_i)\right) f(\lambda_c^*).$$

Inserting (S.7), using P.1, and recalling that $\lambda_i = \lambda_c^*$, we further obtain

$$C' = \left(\prod_{i\,:\,c_i=c} \prod_{k\in\mathcal{K}_i} \mathrm{Pois}(m_{k,i}; \lambda_c^*)\right) \Gamma\big(\lambda_c^*; a_\lambda, b_\lambda\big). \tag{S.38}$$

The product of Poisson distributions in (S.38) can be interpreted as a Poisson likelihood function with unknown rate parameter λ_c^*. Using the fact that the gamma distribution is a conjugate prior for such a likelihood function [7, Section 2.6], it follows that

$$C' \propto \Gamma(\lambda_c^*; a'_{\lambda,c}, b'_{\lambda,c}), \tag{S.39}$$

with parameters

$$a'_{\lambda,c} = a_\lambda + \sum_{i\,:\,c_i=c} M^{(i)}, \quad b'_{\lambda,c} = b_\lambda + N(c). \tag{S.40}$$

S3.4 Final Expression of $f(\boldsymbol{p}_c^*|\boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z})$

Inserting (S.34), (S.36), and (S.39) into (S.32), we finally arrive at

$$\begin{aligned} f(\boldsymbol{p}_c^*|\boldsymbol{X}, \boldsymbol{c}, \boldsymbol{P}_{\neg c}^*, \boldsymbol{Z}) \propto\; & \Gamma^{-1}(q_{c,1}^{*2}; a'_{\mathrm{q},c,1}, b'_{\mathrm{q},c,1})\, \Gamma^{-1}(q_{c,2}^{*2}; a'_{\mathrm{q},c,2}, b'_{\mathrm{q},c,2})\, \Gamma^{-1}(d_{c,1}^{*2}; a'_{\mathrm{d},c,1}, b'_{\mathrm{d},c,1}) \\ & \times \Gamma^{-1}(d_{c,2}^{*2}; a'_{\mathrm{d},c,2}, b'_{\mathrm{d},c,2})\, \Gamma(\lambda_c^*; a'_{\lambda,c}, b'_{\lambda,c}), \end{aligned}$$

with the parameters $a'_{\mathrm{q},c,1}$ etc. given by (S.35), (S.37), and (S.40). This result is identical to (27)–(32).

S4 Conditional pmf $p_i(c;\ell)$

We will derive the expressions (25) and (26) for the conditional pmf $p_i(c;\ell)$ involved in the Gibbs sampler (see Sections IV-C1 and V-A). We recall from (24) that $p_i(c;\ell)$ is the evaluation of $p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z})$ for specific ℓ-dependent arguments. The central observation now is that the paired matrices $(\mathbf{X}_{\mathcal{K}_i,i}, \mathbf{Z}_{\mathcal{K}_i,i})_{i=1}^{I}$ can be interpreted as draws from a DPM, and it is known [8, Eq. (3.6)] that

$$p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z}) = \begin{cases} C\,\frac{1}{\alpha+I-1}\,N_{c_i}(\boldsymbol{c}_{\neg i})\,f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i = \boldsymbol{p}^*_{c_i}), & c_i \in \mathcal{C}_{\neg i}, \\ C\,\frac{\alpha}{\alpha+I-1}\int_{\mathbb{R}^5_+} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)\,f_{\mathrm{B}}(\boldsymbol{p}_i)\,\mathrm{d}\boldsymbol{p}_i, & c_i \notin \mathcal{C}_{\neg i}, \end{cases} \tag{S.41}$$

where C is a normalization constant and $N_{c_i}(\boldsymbol{c}_{\neg i})$ is the number of elements in $\boldsymbol{c}_{\neg i}$ equal to c_i. (Note that the expression for the second case of (S.41) depends on $\boldsymbol{c}_{\neg i}$ and $\boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}$ only via C.) It remains to find expressions of the pdfs involved in (S.41).

S4.1 Conditional pmf $p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z})$ for $c_i \in \mathcal{C}_{\neg i}$

The pdf $f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)$ appearing in the first case ($c_i \in \mathcal{C}_{\neg i}$) in (S.41) can be factored as

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)\,f(\boldsymbol{X}_{\mathcal{K}_i,i}|\boldsymbol{p}_i). \tag{S.42}$$

Using the chain rule, the factor $f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)$ can be expanded as

$$f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i) = f(\boldsymbol{Z}_{k_{\mathrm{S},i},i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} f(\boldsymbol{Z}_{k,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{k_{\mathrm{S},i}:k-1,i}, \boldsymbol{p}_i). \tag{S.43}$$

By B.7, we have $f(\boldsymbol{Z}_{k_{\mathrm{S},i},i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i) = f(\boldsymbol{Z}_{k_{\mathrm{S},i},i}|\boldsymbol{x}_{k_{\mathrm{S},i},i}, \boldsymbol{p}_i)$ and $f(\boldsymbol{Z}_{k,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{k_{\mathrm{S},i}:k-1,i}, \boldsymbol{p}_i) = f(\boldsymbol{Z}_{k,i}|\boldsymbol{x}_{k,i}, \boldsymbol{p}_i)$, and thus expression (S.43) simplifies to

$$f(\boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{p}_i) = \prod_{k\in\mathcal{K}_i} f(\boldsymbol{Z}_{k,i}|\boldsymbol{x}_{k,i}, \boldsymbol{p}_i). \tag{S.44}$$

For the factor $f(\boldsymbol{X}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)$ in (S.42), B.1 implies

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = f(\boldsymbol{X}_{\mathcal{K}_i,i}|q^2_{i,1}, q^2_{i,2}),$$

and using (S.17), we obtain further

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \mathbf{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}) \prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \mathcal{N}(\boldsymbol{x}_{k,i}; \boldsymbol{F}\boldsymbol{x}_{k-1,i}, \boldsymbol{Q}_i). \tag{S.45}$$

We now insert (S.44) and (S.45) into (S.42) and obtain

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i) = \left(\prod_{k\in\mathcal{K}_i} f(\boldsymbol{Z}_{k,i}|\boldsymbol{x}_{k,i}, \boldsymbol{p}_i)\right) \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \mathbf{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}) \prod_{k'=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}} \mathcal{N}(\boldsymbol{x}_{k',i}; \boldsymbol{F}\boldsymbol{x}_{k'-1,i}, \boldsymbol{Q}_i). \tag{S.46}$$

Finally, inserting (S.46) into the first case in (S.41) gives

$$p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z}) = C\,\frac{1}{\alpha+I-1}\,N_{c_i}(\boldsymbol{c}_{\neg i})\left(\prod_{k\in\mathcal{K}_i} f(\boldsymbol{Z}_{k,i}\,|\,\boldsymbol{x}_{k,i}, \boldsymbol{p}_i{=}\boldsymbol{p}^*_{c_i})\right)\mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i};\,\boldsymbol{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}})$$
$$\times\prod_{k'=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}}\mathcal{N}(\boldsymbol{x}_{k',i};\,\boldsymbol{F}\boldsymbol{x}_{k'-1,i}, \boldsymbol{Q}^*_{c_i}),$$

for $c_i \in \mathcal{C}_{\neg i}$. This expression is consistent with (25) (via (24)).

S4.2 Conditional pmf $p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z})$ for $c_i \notin \mathcal{C}_{\neg i}$

To calculate the integral $\int_{\mathbb{R}_+^5} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)\,f_{\mathrm{B}}(\boldsymbol{p}_i)\,\mathrm{d}\boldsymbol{p}_i$ appearing in the second case ($c_i \notin \mathcal{C}_{\neg i}$) in (S.41), we will again use the conjugacy of the prior $f_{\mathrm{B}}(\boldsymbol{p}_i)$. The integrand can be written as

$$f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)\,f_{\mathrm{B}}(\boldsymbol{p}_i) = f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)\,f(\boldsymbol{p}_i) = f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{p}_i),$$

where (19) was used. In (S.14), we expressed $f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}, \boldsymbol{p}_i)$ as the product of three factors A, B, and C, which are given by (S.19), (S.24), and (S.25), respectively. After inserting these expressions, the result can be organized into five factors that depend on $\boldsymbol{p}_i$ only via, respectively, $q_{i,1}^2$, $q_{i,2}^2$, $d_{i,1}^2$, $d_{i,2}^2$, and λ_i. As a consequence, our integral can be rewritten as the product of five one-dimensional integrals and another factor, according to

$$\int_{\mathbb{R}_+^5} f(\boldsymbol{X}_{\mathcal{K}_i,i}, \boldsymbol{Z}_{\mathcal{K}_i,i}|\boldsymbol{p}_i)\,f_{\mathrm{B}}(\boldsymbol{p}_i)\,\mathrm{d}\boldsymbol{p}_i = A_1A_2B_1B_2C_1D, \tag{S.47}$$

where

$$A_1 \triangleq \int_{\mathbb{R}_+}\left(\prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}}\prod_{\kappa=1}^{2}\mathcal{N}(u_{k,i,\kappa};0,q_{i,1}^2)\right)\Gamma^{-1}\big(q_{i,1}^2;a_{\mathrm{q},1},b_{\mathrm{q},1}\big)\,\mathrm{d}q_{i,1}^2, \tag{S.48}$$

$$A_2 \triangleq \int_{\mathbb{R}_+}\left(\prod_{k=k_{\mathrm{S},i}+1}^{k_{\mathrm{E},i}}\prod_{\kappa=1}^{2}\mathcal{N}(u_{k,i,\kappa+2};0,q_{i,2}^2)\right)\Gamma^{-1}\big(q_{i,2}^2;a_{\mathrm{q},2},b_{\mathrm{q},2}\big)\,\mathrm{d}q_{i,2}^2, \tag{S.49}$$

$$A_3 \triangleq \mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i};\,\boldsymbol{0}_{4\times1}, \boldsymbol{\Sigma}_{\mathrm{S}}), \tag{S.50}$$

$$B_1 \triangleq \int_{\mathbb{R}_+}\left(\prod_{k\in\mathcal{K}_i}\prod_{n=1}^{m_{k,i}}\mathcal{N}\big(v_{k,i,1}^{(n)};0,d_{i,1}^2\big)\right)\Gamma^{-1}\big(d_{i,1}^2;a_{\mathrm{d},1},b_{\mathrm{d},1}\big)\,\mathrm{d}d_{i,1}^2, \tag{S.51}$$

$$B_2 \triangleq \int_{\mathbb{R}_+}\left(\prod_{k\in\mathcal{K}_i}\prod_{n=1}^{m_{k,i}}\mathcal{N}\big(v_{k,i,2}^{(n)};0,d_{i,2}^2\big)\right)\Gamma^{-1}\big(d_{i,2}^2;a_{\mathrm{d},2},b_{\mathrm{d},2}\big)\,\mathrm{d}d_{i,2}^2, \tag{S.52}$$

$$C_1 \triangleq \int_{\mathbb{R}_+}\left(\prod_{k\in\mathcal{K}_i}\mathrm{Pois}(m_{k,i};\,\lambda_i)\right)\Gamma\big(\lambda_i;a_\lambda,b_\lambda\big)\,\mathrm{d}\lambda_i. \tag{S.53}$$

Note that, in particular, the factors A_1, A_2, and A_3 are due to the factor A in (S.19), the factors B_1 and B_2 are due to the factor B in (S.24), and the factor C_1 is due to the factor C in (S.25).

Factors A_1, A_2, B_1, and B_2. The integrals A_1, A_2, B_1, and B_2 all have the same structure, being the product of multiple Gaussian pdfs times an inverse gamma pdf. Thus, we first calculate these integrals in a generic way. While for a single Gaussian pdf this would be the classical compound Gaussian construction of a t-distribution, we did not find a similar construction in the literature for this more general setting of multiple Gaussians, and

therefore we provide the details here. In what follows, we consider a generic factor $\tilde{A}$ with observations $y_{k,n}$ for $k \in \tilde{\mathcal{K}} = [\tilde{k}_{\mathrm{S}}, \tilde{k}_{\mathrm{E}}] \subseteq \{1, \ldots, K\}$ and $n = 1, 2, \ldots, \tilde{m}_k$, variance σ^2, shape parameter a, and scale parameter b, i.e.,

$$\tilde{A} \triangleq \int_{\mathbb{R}_+} \left(\prod_{k \in \tilde{\mathcal{K}}} \prod_{n=1}^{\tilde{m}_k} \mathcal{N}(y_{k,n}; 0, \sigma^2) \right) \Gamma^{-1}(\sigma^2; a, b) \, \mathrm{d}\sigma^2.$$

We obtain

$$\begin{aligned} \tilde{A} &= \int_{\mathbb{R}_+} \left(\prod_{k \in \tilde{\mathcal{K}}} \prod_{n=1}^{\tilde{m}_k} \frac{1}{\sqrt{2\pi\sigma^2}} \, \mathrm{e}^{-y_{k,n}^2/(2\sigma^2)} \right) \frac{b^a}{\Gamma(a)} \, (\sigma^2)^{-(a+1)} \, \mathrm{e}^{-b/\sigma^2} \mathrm{d}\sigma^2 \\ &= \frac{1}{(2\pi)^{\tilde{M}/2}} \frac{b^a}{\Gamma(a)} \int_{\mathbb{R}_+} (\sigma^2)^{-(a+\tilde{M}/2+1)} \, \exp\left(-\frac{1}{\sigma^2} \left(b + \frac{1}{2} \sum_{k \in \tilde{\mathcal{K}}} \sum_{n=1}^{\tilde{m}_k} y_{k,n}^2 \right) \right) \mathrm{d}\sigma^2, \end{aligned}$$

with $\tilde{M} \triangleq \sum_{k \in \tilde{\mathcal{K}}} \tilde{m}_k$, or more compactly

$$\tilde{A} = \frac{1}{(2\pi)^{\tilde{M}/2}} \frac{b^a}{\Gamma(a)} \int_{\mathbb{R}_+} (\sigma^2)^{-(a+\tilde{M}/2+1)} \, \mathrm{e}^{-(b+\tilde{y}/2)/\sigma^2} \mathrm{d}\sigma^2, \tag{S.54}$$

where we defined

$$\tilde{y} \triangleq \sum_{k \in \tilde{\mathcal{K}}} \sum_{n=1}^{\tilde{m}_k} y_{k,n}^2. \tag{S.55}$$

Multiplying and dividing the integrand in (S.54) by $\frac{\Gamma(a+\tilde{M}/2)}{(b+\tilde{y}/2)^{a+\tilde{M}/2}}$ gives

$$\begin{aligned} (\sigma^2)^{-(a+\tilde{M}/2+1)} \, \mathrm{e}^{-(b+\tilde{y}/2)/\sigma^2} &= \frac{\Gamma(a+\tilde{M}/2)}{(b+\tilde{y}/2)^{a+\tilde{M}/2}} \frac{(b+\tilde{y}/2)^{a+\tilde{M}/2}}{\Gamma(a+\tilde{M}/2)} \, (\sigma^2)^{-(a+\tilde{M}/2+1)} \, \mathrm{e}^{-(b+\tilde{y}/2)/\sigma^2} \\ &= \frac{\Gamma(a+\tilde{M}/2)}{(b+\tilde{y}/2)^{a+\tilde{M}/2}} \, \Gamma^{-1}(\sigma^2; a+\tilde{M}/2, b+\tilde{y}/2). \end{aligned} \tag{S.56}$$

Inserting (S.56) into (S.54), we obtain

$$\begin{aligned} \tilde{A} &= \frac{1}{(2\pi)^{\tilde{M}/2}} \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+\tilde{M}/2)}{(b+\tilde{y}/2)^{a+\tilde{M}/2}} \int_{\mathbb{R}_+} \Gamma^{-1}(\sigma^2; a+\tilde{M}/2, b+\tilde{y}/2) \, \mathrm{d}\sigma^2 \\ &= \frac{1}{(2\pi)^{\tilde{M}/2}} \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+\tilde{M}/2)}{(b+\tilde{y}/2)^{a+\tilde{M}/2}}, \end{aligned} \tag{S.57}$$

where we used the fact that integrating a pdf over its support yields 1. Using the identity $\frac{b^a}{(b+\tilde{y}/2)^{a+\tilde{M}/2}} = \frac{1}{b^{\tilde{M}/2}} \big(1 + \frac{1}{2b}\tilde{y}\big)^{-(2a+\tilde{M})/2}$ in (S.57) yields further

$$\tilde{A} = \frac{\Gamma(a+\tilde{M}/2)}{\Gamma(a)\,(2\pi b)^{\tilde{M}/2}} \left(1 + \frac{1}{2b}\tilde{y} \right)^{-(2a+\tilde{M})/2}. \tag{S.58}$$

Finally, noting that $\tilde{y}$ in (S.55) can be written as $\|\boldsymbol{y}\|_2^2$ where $\boldsymbol{y} \triangleq \big[\boldsymbol{y}_{\tilde{k}_{\mathrm{S}}}^{\mathrm{T}} \, \boldsymbol{y}_{\tilde{k}_{\mathrm{S}}+1}^{\mathrm{T}} \cdots \, \boldsymbol{y}_{\tilde{k}_{\mathrm{E}}}^{\mathrm{T}}\big]^{\mathrm{T}}$ with $\boldsymbol{y}_k \triangleq [y_{k,1} \; y_{k,2} \; \cdots \; y_{k,\tilde{m}_k}]^{\mathrm{T}}$, expression (S.58) becomes

$$\tilde{A} = \frac{\Gamma(a+\tilde{M}/2)}{\Gamma(a)\,(2\pi b)^{\tilde{M}/2}} \left(1 + \frac{1}{2b} \|\boldsymbol{y}\|_2^2 \right)^{-(2a+\tilde{M})/2} = \mathrm{t}\big(\boldsymbol{y}; 2a, \boldsymbol{0}_{\tilde{M}\times 1}, \tfrac{b}{a}\mathbf{I}_{\tilde{M}}\big), \tag{S.59}$$

where $\mathrm{t}\big(\boldsymbol{y}; \nu, \boldsymbol{\mu}, \boldsymbol{\Sigma}\big)$ denotes the pdf of a multivariate t-distribution [9] with ν degrees of freedom, location parameter vector $\boldsymbol{\mu} \in \mathbb{R}^{\tilde{M}}$, and $\tilde{M} \times \tilde{M}$ scale matrix $\boldsymbol{\Sigma}$.

We can now specialize the result (S.59) to the factors A_1, A_2, B_1, and B_2 in (S.48), (S.49), (S.51), and (S.52), respectively. Let $\boldsymbol{u}_{i,\kappa:\kappa+1} \triangleq \big[\boldsymbol{u}_{k_{\mathrm{S},i}+1,i,\kappa:\kappa+1}^{\mathrm{T}}\ \boldsymbol{u}_{k_{\mathrm{S},i}+2,i,\kappa:\kappa+1}^{\mathrm{T}} \cdots \boldsymbol{u}_{k_{\mathrm{E},i},i,\kappa:\kappa+1}^{\mathrm{T}}\big]^{\mathrm{T}}$ with $\boldsymbol{u}_{k,i,\kappa:\kappa+1} \triangleq [u_{k,i,\kappa}\ u_{k,i,\kappa+1}]^{\mathrm{T}}$ for $\kappa = 1, 3$. Then, A_1 in (S.48) is obtained from (S.59) by setting $\boldsymbol{y} = \boldsymbol{u}_{i,1:2}$, $a = a_{\mathrm{q},1}$, $\tilde{M} = 2(k_{\mathrm{E},i} - k_{\mathrm{S},i})$, and $b = b_{\mathrm{q},1}$:

$$A_1 = \mathrm{t}\big(\boldsymbol{u}_{i,1:2}; 2a_{\mathrm{q},1}, \boldsymbol{0}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})\times 1}, \tfrac{b_{\mathrm{q},1}}{a_{\mathrm{q},1}}\mathbf{I}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})}\big). \tag{S.60}$$

Similarly, A_2 in (S.49) is obtained from (S.59) by setting $\boldsymbol{y} = \boldsymbol{u}_{i,3:4}$, $a = a_{\mathrm{q},2}$, $\tilde{M} = 2(k_{\mathrm{E},i} - k_{\mathrm{S},i})$, and $b = b_{\mathrm{q},2}$:

$$A_2 = \mathrm{t}\big(\boldsymbol{u}_{i,3:4}; 2a_{\mathrm{q},2}, \boldsymbol{0}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})\times 1}, \tfrac{b_{\mathrm{q},2}}{a_{\mathrm{q},2}}\mathbf{I}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})}\big). \tag{S.61}$$

Next, let $\boldsymbol{v}_{\mathcal{K}_i,i,\kappa} \triangleq \big[\boldsymbol{v}_{k_{\mathrm{S},i},i,\kappa}^{\mathrm{T}}\ \boldsymbol{v}_{k_{\mathrm{S},i}+1,i,\kappa}^{\mathrm{T}} \cdots \boldsymbol{v}_{k_{\mathrm{E},i},i,\kappa}^{\mathrm{T}}\big]^{\mathrm{T}}$ with $\boldsymbol{v}_{k,i,\kappa} \triangleq \big[v_{k,i,\kappa}^{(1)}\ v_{k,i,\kappa}^{(2)} \cdots v_{k,i,\kappa}^{(m_{k,i})}\big]^{\mathrm{T}}$ for $\kappa = 1, 2$. Then, B_1 in (S.51) is obtained from (S.59) by setting $\boldsymbol{y} = \boldsymbol{v}_{\mathcal{K}_i,i,1}$, $a = a_{\mathrm{d},1}$, $\tilde{M} = M^{(i)} = \sum_{k\in\mathcal{K}_i} m_{k,i}$, and $b = b_{\mathrm{d},1}$:

$$B_1 = \mathrm{t}\big(\boldsymbol{v}_{\mathcal{K}_i,i,1}; 2a_{\mathrm{d},1}, \boldsymbol{0}_{M^{(i)}\times 1}, \tfrac{b_{\mathrm{d},1}}{a_{\mathrm{d},1}}\mathbf{I}_{M^{(i)}}\big), \tag{S.62}$$

and B_2 in (S.52) is obtained from (S.59) by setting $\boldsymbol{y} = \boldsymbol{v}_{\mathcal{K}_i,i,2}$, $a = a_{\mathrm{d},2}$, $\tilde{M} = M^{(i)}$, and $b = b_{\mathrm{d},2}$:

$$B_2 = \mathrm{t}\big(\boldsymbol{v}_{\mathcal{K}_i,i,2}; 2a_{\mathrm{d},2}, \boldsymbol{0}_{M^{(i)}\times 1}, \tfrac{b_{\mathrm{d},2}}{a_{\mathrm{d},2}}\mathbf{I}_{M^{(i)}}\big). \tag{S.63}$$

Factor C_1. It remains to calculate the factor C_1 in (S.53). We obtain

$$\begin{aligned} C_1 &\triangleq \int_{\mathbb{R}_+} \left(\prod_{k\in\mathcal{K}_i} \mathrm{e}^{-\lambda_i} \frac{\lambda_i^{m_{k,i}}}{m_{k,i}!}\right) \frac{b_\lambda^{a_\lambda}}{\Gamma(a_\lambda)} \lambda_i^{a_\lambda - 1} \mathrm{e}^{-b_\lambda \lambda_i}\, \mathrm{d}\lambda_i \\ &= \frac{b_\lambda^{a_\lambda}}{\Gamma(a_\lambda)\prod_{k\in\mathcal{K}_i} m_{k,i}!} \int_{\mathbb{R}_+} \lambda_i^{a_\lambda + M^{(i)} - 1} \mathrm{e}^{-(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)\lambda_i}\, \mathrm{d}\lambda_i, \end{aligned} \tag{S.64}$$

where $|\mathcal{K}_i| = k_{\mathrm{E},i} - k_{\mathrm{S},i} + 1$ was used. Multiplying and dividing the integrand in (S.64) by $\frac{\Gamma(a_\lambda + M^{(i)})}{(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)^{a_\lambda + M^{(i)}}}$ results in

$$\begin{aligned} &\lambda_i^{a_\lambda + M^{(i)} - 1} \mathrm{e}^{-(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)\lambda_i} \\ &\quad = \frac{\Gamma(a_\lambda + M^{(i)})}{(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)^{a_\lambda + M^{(i)}}} \frac{(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)^{a_\lambda + M^{(i)}}}{\Gamma(a_\lambda + M^{(i)})} \lambda_i^{a_\lambda + M^{(i)} - 1} \mathrm{e}^{-(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)\lambda_i} \\ &\quad = \frac{\Gamma(a_\lambda + M^{(i)})}{(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)^{a_\lambda + M^{(i)}}} \Gamma(\lambda_i; a_\lambda + M^{(i)}, k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda). \end{aligned} \tag{S.65}$$

Inserting (S.65) into (S.64) gives

$$\begin{aligned} C_1 &= \frac{b_\lambda^{a_\lambda}}{\Gamma(a_\lambda)\prod_{k\in\mathcal{K}_i} m_{k,i}!} \frac{\Gamma(a_\lambda + M^{(i)})}{(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)^{a_\lambda + M^{(i)}}} \int_{\mathbb{R}_+} \Gamma(\lambda_i; a_\lambda + M^{(i)}, k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)\, \mathrm{d}\lambda_i \\ &= \frac{b_\lambda^{a_\lambda}}{\Gamma(a_\lambda)\prod_{k\in\mathcal{K}_i} m_{k,i}!} \frac{\Gamma(a_\lambda + M^{(i)})}{(k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda)^{a_\lambda + M^{(i)}}}, \end{aligned}$$

where we again used the fact that integrating a pdf over its support yields 1. Rearranging terms gives

$$\begin{aligned} C_1 &= \frac{\Gamma(a_\lambda + M^{(i)}) \big(\frac{b_\lambda}{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda}\big)^{a_\lambda}}{\Gamma(a_\lambda)} \prod_{k\in\mathcal{K}_i} \frac{\big(\frac{1}{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda}\big)^{m_{k,i}}}{m_{k,i}!} \\ &= \mathrm{NM}\big(\boldsymbol{m}_{\mathcal{K}_i,i}; a_\lambda, \tfrac{1}{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda}\mathbf{1}_{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1\times 1}\big). \end{aligned} \tag{S.66}$$

Here, $\mathrm{NM}(\boldsymbol{m}_{\mathcal{K}_i,i}; a, \boldsymbol{b})$ denotes the pdf of the negative multinomial distribution [10] with parameters a and $\boldsymbol{b} \in \Delta_{k_{\mathrm{E},i}-k_{\mathrm{S},i}}$, where $\Delta_{k_{\mathrm{E},i}-k_{\mathrm{S},i}}$ is the $(k_{\mathrm{E},i}-k_{\mathrm{S},i})$-dimensional simplex.

Final expression of $p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z})$ for $c_i \notin \mathcal{C}_{\neg i}$. Inserting (S.60)–(S.63), (S.66), and (S.50) into (S.47) and the result into the second case in (S.41) gives

$$\begin{aligned} p(c_i\,|\,\boldsymbol{c}_{\neg i}, \boldsymbol{P}^*_{\boldsymbol{c}_{\neg i}}, \boldsymbol{X}, \boldsymbol{Z}) = C\frac{\alpha}{\alpha+I-1}\,&\mathrm{t}\big(\boldsymbol{u}_{i,1:2}; 2a_{\mathrm{q},1}, \boldsymbol{0}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})\times 1}, \tfrac{b_{\mathrm{q},1}}{a_{\mathrm{q},1}}\mathbf{I}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})}\big) \\ &\times \mathrm{t}\big(\boldsymbol{u}_{i,3:4}; 2a_{\mathrm{q},2}, \boldsymbol{0}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})\times 1}, \tfrac{b_{\mathrm{q},2}}{a_{\mathrm{q},2}}\mathbf{I}_{2(k_{\mathrm{E},i}-k_{\mathrm{S},i})}\big) \\ &\times \mathrm{t}\big(\boldsymbol{v}_{\mathcal{K}_i,i,1}; 2a_{\mathrm{d},1}, \boldsymbol{0}_{M^{(i)}\times 1}, \tfrac{b_{\mathrm{d},1}}{a_{\mathrm{d},1}}\mathbf{I}_{M^{(i)}}\big)\,\mathrm{t}\big(\boldsymbol{v}_{\mathcal{K}_i,i,2}; 2a_{\mathrm{d},2}, \boldsymbol{0}_{M^{(i)}\times 1}, \tfrac{b_{\mathrm{d},2}}{a_{\mathrm{d},2}}\mathbf{I}_{M^{(i)}}\big) \\ &\times \mathrm{NM}\big(\boldsymbol{m}_{\mathcal{K}_i,i}; a_\lambda, \tfrac{1}{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1+b_\lambda}\mathbf{1}_{k_{\mathrm{E},i}-k_{\mathrm{S},i}+1\times 1}\big)\,\mathcal{N}(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{0}_{4\times 1}, \boldsymbol{\Sigma}_{\mathrm{S}}), \end{aligned}$$

for $c_i \notin \mathcal{C}_{\neg i}$. This expression is consistent with (26) (via (24)).

S5 Description of the PGAS

This section supplements Section IV-B by providing further details of the PGAS, including closed-form expressions for the distributions involved therein.

S5.1 PGAS Sampling

We recall from Section IV-B that, at sampling step ℓ and considering a specific object index $i \in \{1,\ldots,I\}$, we have available the samples $\boldsymbol{X}^{(\ell-1)}_{\mathcal{K}_i,i}$ and $\boldsymbol{p}^{*(\ell-1)}_{c_i^{(\ell-1)}}$ from the previous sampling step $\ell-1$, and the new sample $\boldsymbol{X}^{(\ell)}_{\mathcal{K}_i,i}$ is drawn by randomly choosing it to be equal to particle $\boldsymbol{X}^{(\ell,r)}_{\mathcal{K}_i,i}$ with probability given by weight $w_i^{(\ell,r)}$, for $r \in \{1,\ldots,R\}$ [11]. Here, the particles $\boldsymbol{X}^{(\ell,r)}_{\mathcal{K}_i,i}$ and weights $w_i^{(\ell,r)} \in [0,1]$ depend on $\boldsymbol{X}^{(\ell-1)}_{\mathcal{K}_i,i}$, $\boldsymbol{p}^{*(\ell-1)}_{c_i^{(\ell-1)}}$, and $\boldsymbol{Z}_{\mathcal{K}_i,i}$, and they are obtained for each $r \in \{1,\ldots,R\}$ as follows. First, time-varying preliminary particles ("pre-particles") $\boldsymbol{\Xi}^{(\ell,k,r)}_{k_{\mathrm{S},i}:k,i} \triangleq \big[\boldsymbol{\xi}^{(\ell,k,r)}_{k',i}\big]^{k}_{k'=k_{\mathrm{S},i}}$ and corresponding "pre-weights" $\nu_i^{(\ell,k,r)} \in [0,1]$ are time-recursively calculated for $k = k_{\mathrm{S},i}, k_{\mathrm{S},i}+1,\ldots,k_{\mathrm{E},i}$ by a particle filtering algorithm (described below). Then, the rth particle and weight are taken to be the final (at time $k = k_{\mathrm{E},i}$) pre-particle and pre-weight, i.e.,

$$\boldsymbol{X}^{(\ell,r)}_{\mathcal{K}_i,i} = \boldsymbol{\Xi}^{(\ell,k_{\mathrm{E},i},r)}_{k_{\mathrm{S},i}:k_{\mathrm{E},i},i}, \quad w_i^{(\ell,r)} = \nu_i^{(\ell,k_{\mathrm{E},i},r)}.$$

Next, we describe the particle filtering algorithm that is used to time-recursively generate the $\boldsymbol{\Xi}^{(\ell,k,r)}_{k_{\mathrm{S},i}:k,i}$ and $\nu_i^{(\ell,k,r)}$ [11].

S5.1.1 Particle Filter—Initialization ($k = k_{\mathrm{S},i}$)

At the first time step of object i's existence, $k_{\mathrm{S},i}$, the pre-particles $\boldsymbol{\xi}^{(\ell,k_{\mathrm{S},i},r)}_{k_{\mathrm{S},i},i}$ for $r = 1,\ldots,R-1$ are sampled from a proposal pdf $g_{\mathrm{init}}\big(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{p}^{*(\ell-1)}_{c_i^{(\ell-1)}}, \boldsymbol{Z}_{k_{\mathrm{S},i},i}\big)$ to be specified in Section S5.2.3. For $r = R$, on the other hand, $\boldsymbol{\xi}^{(\ell,k_{\mathrm{S},i},R)}_{k_{\mathrm{S},i},i}$ is deterministically set equal to $\boldsymbol{x}^{(\ell-1)}_{k_{\mathrm{S},i},i}$, i.e., the first subvector of the sample $\boldsymbol{X}^{(\ell-1)}_{\mathcal{K}_i,i} = \big[\boldsymbol{x}^{(\ell-1)}_{k,i}\big]_{k\in\mathcal{K}_i}$ that was obtained in the previous sampling step $\ell-1$. Furthermore, the pre-weights $\nu_i^{(\ell,k_{\mathrm{S},i},r)}$ are calculated for all $r = 1,\ldots,R$ according to

$$\nu_i^{(\ell,k_{\mathrm{S},i},r)} \propto \frac{h_{\mathrm{init}}\big(\boldsymbol{\xi}^{(\ell,k_{\mathrm{S},i},r)}_{k_{\mathrm{S},i},i}, \boldsymbol{p}^{*(\ell-1)}_{c_i^{(\ell-1)}}, \boldsymbol{Z}_{k_{\mathrm{S},i},i}\big)}{g_{\mathrm{init}}\big(\boldsymbol{\xi}^{(\ell,k_{\mathrm{S},i},r)}_{k_{\mathrm{S},i},i}; \boldsymbol{p}^{*(\ell-1)}_{c_i^{(\ell-1)}}, \boldsymbol{Z}_{k_{\mathrm{S},i},i}\big)}, \tag{S.67}$$

where

$$h_{\text{init}}(\boldsymbol{x}_{k_{\text{S},i},i}, \boldsymbol{p}_i, \boldsymbol{Z}_{k_{\text{S},i},i}) \triangleq f(\boldsymbol{Z}_{k_{\text{S},i},i} | \boldsymbol{x}_{k_{\text{S},i},i}, \boldsymbol{p}_i) f(\boldsymbol{x}_{k_{\text{S},i},i} | \boldsymbol{p}_i),$$

and the proportionality constant in (S.67) is chosen such that $\sum_{r=1}^{R} \nu_i^{(\ell,k_{\text{S},i},r)} = 1$. A closed-form expression for the observation pdf $f(\boldsymbol{Z}_{k,i} | \boldsymbol{x}_{k,i}, \boldsymbol{p}_i)$ was provided in (14). The initial state pdf $f(\boldsymbol{x}_{k_{\text{S},i},i} | \boldsymbol{p}_i)$ will be specified in Section S5.2.1.

S5.1.2 Particle Filter—Time Steps $k = k_{\text{S},i}+1, \ldots, k_{\text{E},i}$

At the subsequent time steps $k = k_{\text{S},i}+1, \ldots, k_{\text{E},i}$, the previous (from time step $k-1$) pre-particles $\boldsymbol{\Xi}_{k_{\text{S},i}:k-1,i}^{(\ell,k-1,r)}$ are first resampled. The resampled pre-particles are denoted $\boldsymbol{\Xi}_{k_{\text{S},i}:k-1,i}^{(\ell,k,r)}$ (now using time index k in the superscript). For each $r \in \{1, \ldots, R\}$, $\boldsymbol{\Xi}_{k_{\text{S},i}:k-1,i}^{(\ell,k,r)}$ is randomly chosen to be $\boldsymbol{\Xi}_{k_{\text{S},i}:k-1,i}^{(\ell,k-1,r')}$ with probability $P_i^{(\ell,k-1,r',r)}$ for all $r' \in \{1, \ldots, R\}$. Here, for $r = 1, \ldots, R-1$, $P_i^{(\ell,k-1,r',r)} = \nu_i^{(\ell,k-1,r')}$, while for $r = R$,

$$P_i^{(\ell,k-1,r',R)} \propto \nu_i^{(\ell,k-1,r')} f_{\mathbf{x}_{k,i}|\mathbf{x}_{k-1,i},\mathbf{p}_i}\big(\boldsymbol{x}_{k,i}^{(\ell-1)} \,\big|\, \boldsymbol{\xi}_{k-1,i}^{(\ell,k-1,r')}, \boldsymbol{p}_{c_i^{(\ell-1)}}^{*(\ell-1)}\big),$$

with the proportionality constant chosen such that $\sum_{r'=1}^{R} P_i^{(\ell,k-1,r',R)} = 1$. The state-transition pdf $f(\boldsymbol{x}_{k,i} | \boldsymbol{x}_{k-1,i}, \boldsymbol{p}_i)$ will be specified in Section S5.2.2.

Then, after the resampling step, we obtain the current rth pre-particle $\boldsymbol{\Xi}_{k_{\text{S},i}:k,i}^{(\ell,k,r)}$ by augmenting the resampled pre-particle $\boldsymbol{\Xi}_{k_{\text{S},i}:k-1,i}^{(\ell,k,r)}$ according to

$$\boldsymbol{\Xi}_{k_{\text{S},i}:k,i}^{(\ell,k,r)} \triangleq \big[\boldsymbol{\Xi}_{k_{\text{S},i}:k-1,i}^{(\ell,k,r)} \;\; \boldsymbol{\xi}_{k,i}^{(\ell,k,r)}\big].$$

Here, for $r = 1, \ldots, R-1$, $\boldsymbol{\xi}_{k,i}^{(\ell,k,r)}$ is sampled from a proposal pdf $g\big(\boldsymbol{x}_{k,i}; \boldsymbol{\xi}_{k-1,i}^{(\ell,k-1,r)}, \boldsymbol{p}_{c_i^{(\ell-1)}}^{*(\ell-1)}, \boldsymbol{Z}_{k,i}\big)$ to be specified in Section S5.2.4. For $r = R$, on the other hand, $\boldsymbol{\xi}_{k,i}^{(\ell,k,R)}$ is deterministically set equal to $\boldsymbol{x}_{k,i}^{(\ell-1)}$, i.e., the relevant subvector of the sample $\boldsymbol{X}_{\mathcal{K}_i,i}^{(\ell-1)} = \big[\boldsymbol{x}_{k,i}^{(\ell-1)}\big]_{k \in \mathcal{K}_i}$ obtained in the previous sampling step $\ell-1$. Furthermore, the pre-weights $\nu_i^{(\ell,k,r)}$ are calculated for all $r = 1, \ldots, R$ according to

$$\nu_i^{(\ell,k,r)} \propto \frac{h\big(\boldsymbol{\xi}_{k,i}^{(\ell,k,r)}, \boldsymbol{\xi}_{k-1,i}^{(\ell,k-1,r)}, \boldsymbol{p}_{c_i^{(\ell-1)}}^{*(\ell-1)}, \boldsymbol{Z}_{k,i}\big)}{g\big(\boldsymbol{\xi}_{k,i}^{(\ell,k,r)}; \boldsymbol{\xi}_{k-1,i}^{(\ell,k-1,r)}, \boldsymbol{p}_{c_i^{(\ell-1)}}^{*(\ell-1)}, \boldsymbol{Z}_{k,i}\big)}, \tag{S.68}$$

where

$$h(\boldsymbol{x}_{k,i}, \boldsymbol{x}_{k-1,i}, \boldsymbol{p}_i, \boldsymbol{Z}_{k,i}) \triangleq f(\boldsymbol{Z}_{k,i} | \boldsymbol{x}_{k,i}, \boldsymbol{p}_i) f(\boldsymbol{x}_{k,i} | \boldsymbol{x}_{k-1,i}, \boldsymbol{p}_i),$$

and the proportionality constant in (S.68) is chosen such that $\sum_{r=1}^{R} \nu_i^{(\ell,k,r)} = 1$.

S5.2 Distributions Involved in the PGAS

According to Section S5.1, the distributions involved in the PGAS include the initial state pdf $f(\boldsymbol{x}_{k_{\text{S},i},i} | \boldsymbol{p}_i)$, the state-transition pdf $f(\boldsymbol{x}_{k,i} | \boldsymbol{x}_{k-1,i}, \boldsymbol{p}_i)$, and the proposal pdfs $g_{\text{init}}(\boldsymbol{x}_{k_{\text{S},i},i}; \boldsymbol{p}_{c_i}^*, \boldsymbol{Z}_{k_{\text{S},i},i})$ and $g(\boldsymbol{x}_{k,i}; \boldsymbol{x}_{k-1,i}, \boldsymbol{p}_{c_i}^*, \boldsymbol{Z}_{k,i})$.

S5.2.1 Initial State pdf

Using B.6, the initial state pdf $f(\boldsymbol{x}_{k_{\text{S},i},i} | \boldsymbol{p}_i)$ reduces to $f(\boldsymbol{x}_{k_{\text{S},i},i})$ in (1). Thus, we have

$$f(\boldsymbol{x}_{k_{\text{S},i},i} | \boldsymbol{p}_i) = \mathcal{N}(\boldsymbol{x}_{k_{\text{S},i},i}; \mathbf{0}_{4\times 1}, \boldsymbol{\Sigma}_{\text{S}}),$$

with some known initial state covariance matrix $\boldsymbol{\Sigma}_{\text{S}} \in \mathbb{R}^{4\times 4}$.

S5.2.2 State-Transition pdf

Using B.5, the state-transition pdf $f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i},\boldsymbol{p}_i) = f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i},\boldsymbol{Q}_i,\boldsymbol{D}_i,\lambda_i)$ reduces to $f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i},\boldsymbol{Q}_i)$ in (4). Thus, we have

$$f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i},\boldsymbol{p}_i) = \mathcal{N}(\boldsymbol{x}_{k,i};\boldsymbol{F}\boldsymbol{x}_{k-1,i},\boldsymbol{Q}_i)\,.$$

S5.2.3 Proposal pdf $g_{\text{init}}(\boldsymbol{x}_{k_{\text{S},i},i};\boldsymbol{p}^*_{c_i},\boldsymbol{Z}_{k_{\text{S},i},i})$

The object state proposal pdf used in the initialization of the PGAS particle filter is chosen as a Gaussian pdf, i.e.,

$$g_{\text{init}}(\boldsymbol{x}_{k_{\text{S},i},i};\boldsymbol{p}^*_{c_i},\boldsymbol{Z}_{k_{\text{S},i},i}) = \mathcal{N}\big(\boldsymbol{x}_{k_{\text{S},i},i};\boldsymbol{\mu}^{(g_{\text{init}})}_{k_{\text{S},i},i},\boldsymbol{\Sigma}^{(g_{\text{init}})}_{k_{\text{S},i},i}\big)\,.$$

Here,

$$\boldsymbol{\mu}^{(g_{\text{init}})}_{k_{\text{S},i},i} = [\bar{\boldsymbol{z}}^{\text{T}}_{k_{\text{S},i},i}\ 0\ 0]^{\text{T}},$$

with $\bar{\boldsymbol{z}}_{k,i}$ as defined in (21), and

$$\boldsymbol{\Sigma}^{(g_{\text{init}})}_{k_{\text{S},i},i} = \frac{\overline{d^{*2}_{c_i}}}{m_{k_{\text{S},i},i}}\,\text{diag}(\mathbf{I}_2,\mathbf{0}_{2\times 2}) + \boldsymbol{\Sigma}_{\text{S}}\circ\text{diag}(\mathbf{0}_{2\times 2},\mathbf{I}_2)\,,$$

where $\circ$ denotes the Hadamard/elementwise product and $\overline{d^{*2}_c} \triangleq \frac{d^{*2}_{c,1}+d^{*2}_{c,2}}{2}$. These expressions of $\boldsymbol{\mu}^{(g_{\text{init}})}_{k_{\text{S},i},i}$ and $\boldsymbol{\Sigma}^{(g_{\text{init}})}_{k_{\text{S},i},i}$ combine prior and empirical information. More specifically, $\boldsymbol{\mu}^{(g_{\text{init}})}_{k_{\text{S},i},i}$ is the sum of the prior mean of $f(\boldsymbol{x}_{k_{\text{S},i},i})$ (which is $\mathbf{0}_{4\times 1}$ by (1)) and the average of the measurements associated with object i at time $k_{\text{S},i}$, $\bar{\boldsymbol{z}}_{k_{\text{S},i},i}$. This is inspired by the MMSE estimator for the sequential linear Gaussian case [12, Sec. 12.6], which can be decomposed into analogous prior and empirical terms. Furthermore, the two position-related diagonal elements of $\boldsymbol{\Sigma}^{(g_{\text{init}})}_{k_{\text{S},i},i}$ are given by $\overline{d^{*2}_{c_i}}/m_{k_{\text{S},i},i}$, i.e., the average of the two object extent parameters of object i, $\overline{d^{*2}_{c_i}}$, divided by the number of associated measurements, $m_{k_{\text{S},i},i}$. Thus, the position elements of the initial state $\boldsymbol{x}_{k_{\text{S},i},i}$ are more tightly distributed around the position elements of the mean $\boldsymbol{\mu}^{(g_{\text{init}})}_{k_{\text{S},i},i}$ when $\overline{d^{*2}_{c_i}}$ is smaller and/or $m_{k_{\text{S},i},i}$ is larger. Finally, the velocity-related diagonal elements of $\boldsymbol{\Sigma}^{(g_{\text{init}})}_{k_{\text{S},i},i}$ equal the corresponding elements of the prior covariance matrix $\boldsymbol{\Sigma}_{\text{S}}$ in (1), since no velocity measurements are available.

S5.2.4 Proposal pdf $g(\boldsymbol{x}_{k,i};\boldsymbol{x}_{k-1,i},\boldsymbol{p}^*_{c_i},\boldsymbol{Z}_{k,i})$

The object state proposal pdf $g(\boldsymbol{x}_{k,i};\boldsymbol{x}_{k-1,i},\boldsymbol{p}^*_{c_i},\boldsymbol{Z}_{k,i})$ used by the particle filter for $k = k_{\text{S},i}+1,\ldots,k_{\text{E},i}$ is chosen as

$$g(\boldsymbol{x}_{k,i};\boldsymbol{x}_{k-1,i},\boldsymbol{p}^*_{c_i},\boldsymbol{Z}_{k,i}) = \mathcal{N}\big(\boldsymbol{x}_{k,i};\boldsymbol{\mu}^{(g)}_{k,i},\boldsymbol{\Sigma}^{(g)}_{k,i}\big)\,,$$

where

$$\boldsymbol{\mu}^{(g)}_{k,i} = \boldsymbol{F}\boldsymbol{x}_{k-1,i} + \rho_{k,i}\boldsymbol{H}^{\text{T}}(\bar{\boldsymbol{z}}_{k,i} - \boldsymbol{F}\boldsymbol{x}_{k-1,i}) \tag{S.69}$$

and

$$\boldsymbol{\Sigma}^{(g)}_{k,i} = \boldsymbol{Q}^*_{c_i} - q^{*2}_{c_i,1}\,\rho_{k,i}\,\text{diag}(\mathbf{I}_2,\mathbf{0}_{2\times 2})\,, \tag{S.70}$$

with $\rho_{k,i} \triangleq q^{*2}_{c_i,1}/\big(q^{*2}_{c_i,1} + \overline{d^{*2}_{c_i}}/m_{k,i}\big)$. These expressions of $\boldsymbol{\mu}^{(g)}_{k,i}$ and $\boldsymbol{\Sigma}^{(g)}_{k,i}$ again combine prior and empirical information and are inspired by the MMSE estimator for the sequential linear Gaussian case [12, Sec. 12.6]. More specifically, the first term on the right-hand side of (S.69) equals the mean of $f(\boldsymbol{x}_{k,i}|\boldsymbol{x}_{k-1,i},\boldsymbol{Q}_i)$ in (4) while the second term is a correction term involving the average of the measurements associated with object i, $\bar{\boldsymbol{z}}_{k,i}$ defined in (21), and a scaling factor $\rho_{k,i}$ that depends on $q^{*2}_{c_i,1}$, $\overline{d^{*2}_{c_i}}$, and $m_{k,i}$. The correction term—and, thus, the influence

Algorithm S1 Proposed MCMC algorithm (iteration ℓ)

Input: $\boldsymbol{X}^{(\ell-1)}, \boldsymbol{P}^{(\ell-1)} = \big[\boldsymbol{p}_i^{(\ell-1)}\big]_{i=1}^{I}, \boldsymbol{c}^{(\ell-1)}, \big[\boldsymbol{Z}_{k,0:I}\big]_{k=1}^{K}, \big[\mathcal{K}_i\big]_{i=1}^{I}$

1: **for all** $i = 1, \ldots, I$ **do**

2: generate $\boldsymbol{X}_{\mathcal{K}_i,i}^{(\ell)}$ using Algorithm S2 with input $\boldsymbol{X}_{\mathcal{K}_i,i}^{(\ell-1)}, \boldsymbol{p}_i^{(\ell-1)}, \boldsymbol{Z}_{\mathcal{K}_i,i} = [\boldsymbol{Z}_{k,i}]_{k\in\mathcal{K}_i}$

3: **end for**

4: generate $\boldsymbol{P}^{(\ell)}$ and $\boldsymbol{c}^{(\ell)}$ using Algorithm S3 with input $\boldsymbol{X}^{(\ell)}, \boldsymbol{P}^{(\ell-1)}, \boldsymbol{c}^{(\ell-1)}, \boldsymbol{Z} = \big[\boldsymbol{Z}_{k,0:I}\big]_{k=1}^{K}$

Output: $\boldsymbol{X}^{(\ell)}, \boldsymbol{P}^{(\ell)}, \boldsymbol{c}^{(\ell)}$

of the empirical information provided by $\bar{\boldsymbol{z}}_{k,i}$ on $\boldsymbol{\mu}_{k,i}^{(g)}$—tends to be larger when $q_{c_i,1}^{*2}$ is larger and/or $\overline{d_{c_i}^{*2}}$ is smaller and/or $m_{k,i}$ is larger. Furthermore, $\boldsymbol{\Sigma}_{k,i}^{(g)}$ in (S.70) equals the covariance matrix in (4) (via the identity $\boldsymbol{Q}_{c_i}^{*} = \boldsymbol{Q}_i$) except that $q_{c_i,1}^{*2}\,\rho_{k,i}$ is subtracted from the position-related diagonal elements. Thus, the position elements of $\boldsymbol{x}_{k,i}$ are more tightly distributed around the position elements of the mean $\boldsymbol{\mu}_{k,i}^{(g)}$ when $\rho_{k,i}$ is larger. Note that $\boldsymbol{\Sigma}_{k,i}^{(g)}$ is guaranteed to be positive definite because $q_{c_i,1}^{*2}\rho_{k,i} = q_{c_i,1}^{*4}\big/\big(q_{c_i,1}^{*2} + \overline{d_{c_i}^{*2}}/m_{k,i}\big) \in (0, q_{c_i,1}^{*2})$ and thus the subtraction (S.70) does not result in the diagonal elements of the diagonal matrix $\boldsymbol{\Sigma}_{k,i}^{(g)}$ being less than or equal to zero. The construction $\boldsymbol{H}^{\mathrm{T}}(\bar{\boldsymbol{z}}_{k,i} - \boldsymbol{F}\boldsymbol{x}_{k-1,i})$ in (S.69) is motivated by the state-evolution model (2) and measurement model (6).

S6 Pseudo-code of the MCMC Algorithm

Pseudo-code for one iteration of the proposed MCMC algorithm described in Section IV is presented in Algorithms S1–S3. According to Algorithm S1, the MCMC algorithm consists of sampling the object states (using the PGAS stated in Algorithm S2) and then sampling the class indices and the class parameters/object parameters (using the Gibbs sampler stated in Algorithm S3). We note that the DA-aware measurement matrices $\boldsymbol{Z}_{k,0:I}$ for each $k = 1, \ldots, K$ used as inputs to Algorithm S1 are obtained according to $\boldsymbol{Z}_{k,0:I} = \pi_k^{-1}(\boldsymbol{Z}_k)$. Furthermore, the new class index value $c_{i,\text{new}}^{(\ell)}$ involved in Algorithm S3 is chosen as the smallest integer that is larger than the maximum class index value that arose in the previous sampling steps $\ell' = 1, \ldots, \ell-1$ or in the current sampling step ℓ so far (for $i' = 1, \ldots, i-1$), i.e.,

$$c_{i,\text{new}}^{(\ell)} \triangleq \max\big\{c_{\max}^{(\ell-1)}, \max_{i'\in\{1,\ldots,i-1\}} c_{i'}^{(\ell)}\big\} + 1, \tag{S.71}$$

with $c_{\max}^{(\ell-1)} \triangleq \max_{\ell'\in\{1,\ldots,\ell-1\}, i\in\{1,\ldots,I\}} c_i^{(\ell')}$.

MATLAB code for our algorithm is provided at `https://github.com/tjbucco/EMT-BNP-Learning/`.

S7 Further Information on the Overall EMT Method

S7.1 Blackwell-MacQueen Sampling Procedure

According to Section VI-A, in the case where $\hat{I}^{(j-1)} < I_{\max}$, the DAOD parameters $\boldsymbol{p}_{i,\text{DAOD}}^{(j)}$ for $i = \hat{I}^{(j-1)} + 1, \ldots, I_{\max}$ required by the DAOD algorithm are sampled using the Blackwell-MacQueen construction of the DP [13]. This sampling procedure is described in the following.

Algorithm S2 PGAS: sampling the object states (iteration ℓ)

Input: $\boldsymbol{X}_{\mathcal{K}_i,i}^{(\ell-1)}$, $\boldsymbol{p}_i^{(\ell-1)}$, $\boldsymbol{Z}_{\mathcal{K}_i,i} = [\boldsymbol{Z}_{k,i}]_{k\in\mathcal{K}_i}$

for all $r = 1,\ldots,R-1$ **do**
  generate $\boldsymbol{\xi}_{k_{\mathrm{S},i},i}^{(\ell,k_{\mathrm{S},i},r)}$ from $g_{\text{init}}\big(\boldsymbol{x}_{k_{\mathrm{S},i},i}; \boldsymbol{p}_i^{(\ell-1)}, \boldsymbol{Z}_{k_{\mathrm{S},i},i}\big)$ (see Section S5.2.3)
  set $\boldsymbol{\Xi}_{k_{\mathrm{S},i},i}^{(\ell,k_{\mathrm{S},i},r)} = \boldsymbol{\xi}_{k_{\mathrm{S},i},i}^{(\ell,k_{\mathrm{S},i},r)}$
end for
set $\boldsymbol{\xi}_{k_{\mathrm{S},i},i}^{(\ell,k_{\mathrm{S},i},R)} = \boldsymbol{x}_{k_{\mathrm{S},i},i}^{(\ell-1)}$
set $\boldsymbol{\Xi}_{k_{\mathrm{S},i},i}^{(\ell,k_{\mathrm{S},i},R)} = \boldsymbol{\xi}_{k_{\mathrm{S},i},i}^{(\ell,k_{\mathrm{S},i},R)}$
for all $r = 1,\ldots,R$ **do**
  calculate $\nu_i^{(\ell,k_{\mathrm{S},i},r)}$ according to (S.67)
end for
for all $k = k_{\mathrm{S},i}+1,\ldots,k_{\mathrm{E},i}$ **do**
  for all $r = 1,\ldots,R-1$ **do**
    generate $\boldsymbol{\Xi}_{k_{\mathrm{S},i}:k-1,i}^{(\ell,k,r)} = \boldsymbol{\Xi}_{k_{\mathrm{S},i}:k-1,i}^{(\ell,k-1,r')}$ with probability $\nu_i^{(\ell,k-1,r')}$, $r' \in \{1,\ldots,R\}$
    generate $\boldsymbol{\xi}_{k,i}^{(\ell,k,r)}$ from $g\big(\boldsymbol{x}_{k,i}; \boldsymbol{\xi}_{k-1,i}^{(\ell,k-1,r)}, \boldsymbol{p}_i^{(\ell-1)}, \boldsymbol{Z}_{k,i}\big)$ (see Section S5.2.4)
    set $\boldsymbol{\Xi}_{k_{\mathrm{S},i}:k,i}^{(\ell,k,r)} = \big[\boldsymbol{\Xi}_{k_{\mathrm{S},i}:k-1,i}^{(\ell,k,r)} \;\; \boldsymbol{\xi}_{k,i}^{(\ell,k,r)}\big]$
  end for
  for all $r = 1,\ldots,R$ **do**
    set $\tilde{P}_i^{(\ell,k-1,r,R)} = \nu_i^{(\ell,k-1,r)}\, f_{\mathbf{x}_{k,i}|\mathbf{x}_{k-1,i},\mathbf{p}_i}\big(\boldsymbol{x}_{k,i}^{(\ell-1)} \,\big|\, \boldsymbol{\xi}_{k-1,i}^{(\ell,k-1,r)}, \boldsymbol{p}_i^{(\ell-1)}\big)$
  end for
  for all $r = 1,\ldots,R$ **do**
    set $P_i^{(\ell,k-1,r,R)} = \frac{\tilde{P}_i^{(\ell,k-1,r,R)}}{\sum_{r'=1}^{R} \tilde{P}_i^{(\ell,k-1,r',R)}}$
  end for
  generate $\boldsymbol{\Xi}_{k_{\mathrm{S},i}:k-1,i}^{(\ell,k,R)} = \boldsymbol{\Xi}_{k_{\mathrm{S},i}:k-1,i}^{(\ell,k-1,r')}$ with probability $P_i^{(\ell,k-1,r',R)}$, $r' \in \{1,\ldots,R\}$
  set $\boldsymbol{\Xi}_{k_{\mathrm{S},i}:k,i}^{(\ell,k,R)} = \big[\boldsymbol{\Xi}_{k_{\mathrm{S},i}:k-1,i}^{(\ell,k,R)} \;\; \boldsymbol{x}_{k,i}^{(\ell-1)}\big]$
  for all $r = 1,\ldots,R$ **do**
    calculate $\nu_i^{(\ell,k,r)}$ according to (S.68)
  end for
end for
generate $\boldsymbol{X}_{\mathcal{K}_i,i}^{(\ell)} = \boldsymbol{\Xi}_{k_{\mathrm{S},i}:k_{\mathrm{E},i},i}^{(\ell,k_{\mathrm{E},i},r)}$ with probability $\nu_i^{(\ell,k_{\mathrm{E},i},r)}$, $r \in \{1,\ldots,R\}$

Output: $\boldsymbol{X}_{\mathcal{K}_i,i}^{(\ell)}$

First, class index values $c_{i,\text{DAOD}}^{(j)} \in \mathbb{N}$ are generated for $i = \hat{I}^{(j-1)}+1,\ldots,I_{\max}$. More specifically, $c_{i,\text{DAOD}}^{(j)}$ is recursively sampled from the conditional pmf [14, Sec. 2.6.6]

$$p\big(c_{i,\text{DAOD}}^{(j)} \big| c_{1,\text{DAOD}}^{(j)},\ldots,c_{i-1,\text{DAOD}}^{(j)}\big) = \frac{1}{\alpha+i-1}\Bigg(\alpha\,\mathbb{1}\big(c_{i,\text{DAOD}}^{(j)} = c_{i-1,\text{DAOD,max}}^{(j)}+1\big) + \sum_{c=1}^{c_{i-1,\text{DAOD,max}}^{(j)}} N\big(c; c_{1,\text{DAOD}}^{(j)},\ldots,c_{i-1,\text{DAOD}}^{(j)}\big)\,\mathbb{1}\big(c = c_{i,\text{DAOD}}^{(j)}\big)\Bigg),$$

Algorithm S3 Gibbs sampler: sampling the class indices and class/object parameters (iteration ℓ)

Input: $\boldsymbol{X}^{(\ell)}, \boldsymbol{P}^{(\ell-1)} = \big[\boldsymbol{p}_i^{(\ell-1)}\big]_{i=1}^{I}, \boldsymbol{c}^{(\ell-1)}, \boldsymbol{Z} = \big[\boldsymbol{Z}_{k,0:I}\big]_{k=1}^{K}$

for all $i = 1, \ldots, I$ **do**
  set $\boldsymbol{p}^{*(\ell-1)}_{c_i^{(\ell-1)}} = \boldsymbol{p}_i^{(\ell-1)}$ (cf. (18))
end for
for all $i = 1, \ldots, I$ **do**
  set $c_{i,\text{new}}^{(\ell)}$ according to (S.71)
  generate $c_i^{(\ell)} \in \mathcal{C}_{\neg i}^{(\ell,\ell-1)} \cup \{c_{i,\text{new}}^{(\ell)}\}$ from $p_i(c;\ell)$ (see (25) for $c \in \mathcal{C}_{\neg i}^{(\ell,\ell-1)}$ and (26) for $c = c_{i,\text{new}}^{(\ell)}$)
  if $c_i^{(\ell)} = c_{i,\text{new}}^{(\ell)}$
    generate $\boldsymbol{p}^{*(\ell-1)}_{c_{i,\text{new}}^{(\ell)}}$ from $f_{\mathbf{p}_c^*|\mathbf{X},\mathbf{c},\mathbf{P}_{\neg c}^*,\mathbf{Z}}\big(\boldsymbol{p}_c^* \,\big|\, \boldsymbol{X}^{(\ell)}, \boldsymbol{c}^{(\ell)}, \boldsymbol{P}^{*(\ell-1)}_{\boldsymbol{c}_{\neg i}^{(\ell,\ell-1)}}, \boldsymbol{Z}\big)$ (see Section IV-C2 and (27))
end for
for all $c \in \big\{c_1^{(\ell)}, \ldots, c_I^{(\ell)}\big\}$ **do**
  generate $\boldsymbol{p}_c^{*(\ell)}$ from $f_{\mathbf{p}_c^*|\mathbf{X},\mathbf{c},\mathbf{P}_{\neg c}^*,\mathbf{Z}}\big(\boldsymbol{p}_c^* \,\big|\, \boldsymbol{X}^{(\ell)}, \boldsymbol{c}^{(\ell)}, \boldsymbol{P}_{\neg c}^{*(\ell,\ell-1)}, \boldsymbol{Z}\big)$ (see Section IV-C2 and (27))
end for
for all $i = 1, \ldots, I$ **do**
  set $\boldsymbol{p}_i^{(\ell)} = \boldsymbol{p}^{*(\ell)}_{c_i^{(\ell)}}$ (cf. (18))
end for

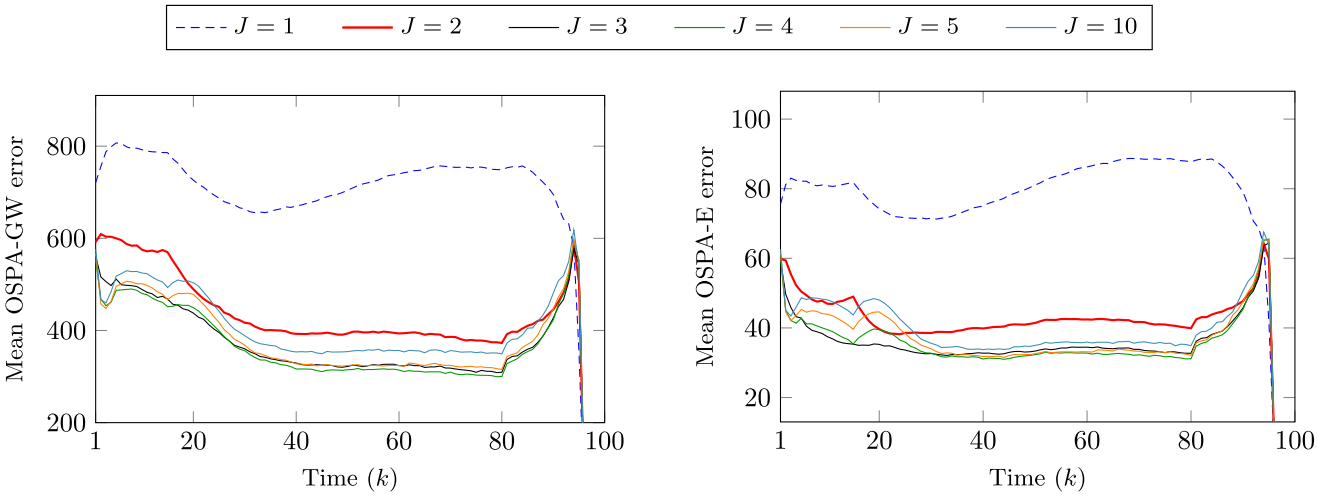
Output: $\boldsymbol{P}^{(\ell)}, \boldsymbol{c}^{(\ell)}$

for $i = \hat{I}^{(j-1)} + 1, \ldots, I_{\max}$. Here, the $c_{i,\text{DAOD}}^{(j)}$ for $i = 1, \ldots, \hat{I}^{(j-1)}$ equal the respective estimates $\hat{c}_i^{(j-1)}$ that were previously calculated by the MCMC estimation algorithm; $\mathbf{1}(\cdot)$ is 1 if the function argument is true and 0 otherwise; $c_{i-1,\text{DAOD},\max}^{(j)} \triangleq \max_{i' \in \{1,\ldots,i-1\}} c_{i',\text{DAOD}}^{(j)}$; and $N\big(c; c_{1,\text{DAOD}}^{(j)}, \ldots, c_{i-1,\text{DAOD}}^{(j)}\big)$ is the number of $c_{i',\text{DAOD}}^{(j)}$ with $i' \in \{1, \ldots, i-1\}$ that are equal to c. We note that the above pmf is related to the Chinese restaurant process, which is a Bayesian nonparametric model closely related to the DP [14, Sec. 3.1].

The DAOD parameters $\boldsymbol{p}_{i,\text{DAOD}}^{(j)}$ for $i = \hat{I}^{(j-1)} + 1, \ldots, I_{\max}$ are now obtained as follows. If $c_{i,\text{DAOD}}^{(j)} \in \big\{c_{1,\text{DAOD}}^{(j)}, \ldots, c_{\hat{I}^{(j-1)},\text{DAOD}}^{(j)}\big\}$, i.e., $c_{i,\text{DAOD}}^{(j)}$ is the index of a class for which the MCMC estimation algorithm already calculated a parameter estimate $\hat{\boldsymbol{p}}^{*(j-1)}_{c_{i,\text{DAOD}}^{(j)}}$, then $\boldsymbol{p}_{i,\text{DAOD}}^{(j)} = \hat{\boldsymbol{p}}^{*(j-1)}_{c_{i,\text{DAOD}}^{(j)}}$. In the opposite case, i.e., $c_{i,\text{DAOD}}^{(j)} \notin \big\{c_{1,\text{DAOD}}^{(j)}, \ldots, c_{\hat{I}^{(j-1)},\text{DAOD}}^{(j)}\big\}$, there are two subcases: (i) If $c_{i,\text{DAOD}}^{(j)} = c_{i-1,\text{DAOD},\max}^{(j)} + 1$, then $\boldsymbol{p}_{i,\text{DAOD}}^{(j)}$ is sampled from the base pdf f_{B} in (20); (ii) if $c_{i,\text{DAOD}}^{(j)} \in \big\{\hat{c}_{\max}^{(j-1)} + 1, \ldots, c_{i-1,\text{DAOD},\max}^{(j)}\big\}$ with $\hat{c}_{\max}^{(j-1)} \triangleq \max_{i \in \{1,\ldots,\hat{I}^{(j-1)}\}} \hat{c}_i^{(j-1)}$, i.e., $c_{i,\text{DAOD}}^{(j)}$ is the index of a class for which the DAOD parameter was already sampled in the former subcase, then that sample is used.

S7.2 Further Details of the DAOD Algorithm

As described in Section VI-B, the DAOD algorithm employs an EM clustering method in both the DA and object existence detection parts in order to account for the fact that an object can generate multiple measurements. In the DA part, the EM clustering method is used to cluster the measurements $\boldsymbol{z}_k^{(\nu)}$. The number of clusters is $\hat{I}_{k-1}^{(j)} + 1$, where $\hat{I}_{k-1}^{(j)}$ for $k \geq 2$ (not to be confused with $\hat{I}^{(j)}$) is an internal estimate of the number of objects at time $k-1$ that is calculated by the object existence detection part of [15], and $\hat{I}_0^{(j)} = 0$. The centers of the first $\hat{I}_{k-1}^{(j)}$ clusters



(a) (b)

Figure S2: Performance of PRO for different numbers of iterations J: (a) Mean OSPA-GW error, (b) mean OSPA-E error.

are then used as measurements at the input of the DA part of [15]. To each of these $\hat{I}_{k-1}^{(j)}$ clusters, the DA part of [15] associates one or none of the POs. Finally, DA is achieved by associating each measurement $\boldsymbol{z}_k^{(\nu)}$ to the PO to which the cluster containing $\boldsymbol{z}_k^{(\nu)}$ was associated by the DA part of [15]. On the other hand, measurements $\boldsymbol{z}_k^{(\nu)}$ assigned to the $(\hat{I}_{k-1}^{(j)}+1)$th cluster by the EM clustering method are considered to be clutter.

In the object existence detection part of the DAOD algorithm, the EM clustering method is used to cluster the clutter measurements. The number of clusters is here chosen as $\varkappa+1$, where $\varkappa$ is determined by a model selection stage that performs EM-based clustering for $\varkappa' = 1, \ldots, \varkappa_{\text{max}}$ clusters and then selects the optimal value of $\varkappa'$ according to the Akaike information criterion [16, 17]. Then, $\varkappa$ new POs are created, whose initial positions are set to the centers of the first $\varkappa$ clusters. These new POs are equipped with object parameters $\boldsymbol{p}_i$ that are obtained by sampling without replacement $\varkappa$ times from the subset of those DAOD parameters $\boldsymbol{p}_{i,\text{DAOD}}^{(j)}$ to which no object is currently associated. The new POs and their object parameters constitute the output of the object existence detection part of the DAOD algorithm.

S7.3 Dependence of Performance on the Number of Iterations

As mentioned in Section VII-A, the simulation results shown in Sections VII-B through VII-D were obtained by carrying out $J=2$ iterations of the MCMC algorithm or the overall EMT method. To justify this choice, we now demonstrate the tracking and parameter estimation performance of PRO (overall EMT method) for other choices of the number of iterations J. Fig. S2 shows the mean OSPA-GW and OSPA-E errors, obtained by averaging over 1000 datasets, for $J \in \{1, 2, 3, 4, 5, 10\}$. Based on these results, it can be argued that with $J=2$ iterations a good tracking and parameter estimation performance is obtained while keeping the computational complexity low. Furthermore, these results show that the mean OSPA-GW and OSPA-E errors do not consistently decrease with growing J; the smallest errors are obtained for $J=4$.

S8 Further Details of the Simulation Setup

Finally, we provide parameters of our simulation and of the considered methods that are not specified in the main paper because of space limitations.

The initial positions of the 15 objects are regularly spaced on a circle of radius $100\,\mathrm{m}$, and the objects move with an initial velocity of $10\,\mathrm{m/s}$ toward the center of the circle. The mean driving noise variances are $\mathbb{E}[\mathsf{q}_{c,1}^{*2}] = 110\,\mathrm{m}^2$ and $\mathbb{E}[\mathsf{q}_{c,2}^{*2}] = 65\,\mathrm{m}^2/\mathrm{s}^2$, the mean extent parameters (measurement noise variances) are $\mathbb{E}[\mathsf{d}_{c,1}^{*2}] = 400\,\mathrm{m}^2$ and $\mathbb{E}[\mathsf{d}_{c,2}^{*2}] = 250\,\mathrm{m}^2$, and the mean measurement rate is $\mathbb{E}[\lambda_c] = 5$.

SEP uses values of some of the hyperparameters that differ from our system model because this resulted in better performance. Concretely, SEP uses $a_{\mathrm{d},1} = a_{\mathrm{d},2} = 70$, $b_{\mathrm{d},1} = 27600\,\mathrm{m}^2$, and $b_{\mathrm{d},2} = 17250\,\mathrm{m}^2$, which causes the variances of the extent parameters $\mathsf{d}_{i,1}^2$ and $\mathsf{d}_{i,2}^2$ to be significantly smaller than those of the system model (note, however, that the mean extent parameters $\mathbb{E}[\mathsf{d}_{i,1}^2]$ and $\mathbb{E}[\mathsf{d}_{i,2}^2]$ are still equal to those of the system model).

T-PMB uses the fixed driving noise covariance matrices $\boldsymbol{Q}_i = \mathrm{diag}\big(\mathbb{E}[\mathsf{q}_{c,1}^{*2}]\,\mathbf{I}_2, \mathbb{E}[\mathsf{q}_{c,2}^{*2}]\,\mathbf{I}_2\big) = \mathrm{diag}(110\,\mathrm{m}^2\,\mathbf{I}_2, 65\,\mathrm{m}^2/\mathrm{s}^2\,\mathbf{I}_2)$ and the fixed measurement rates $\lambda_i = \mathbb{E}[\lambda_c] = 5$, for all $i = 1, \ldots, I$. Furthermore, T-PMB assumes that the object extent matrix $\mathbf{D}_i$ is time-varying and, at each time step, distributed according to an inverse Wishart distribution [18]. We choose the scale matrix and the number of degrees of freedom of the inverse Wishart distribution as $\mathrm{diag}(b_{\mathrm{d},1}, b_{\mathrm{d},2}) = \mathrm{diag}(6000\,\mathrm{m}^2, 3750\,\mathrm{m}^2)$ and $a_{\mathrm{d},1} + 3 = 19$, respectively. This means that the mean of the object extent matrix is the same as the mean object extent matrix of our model. The remaining parameters for T-PMB are set as in [18].

The parameters of the OSPA-GW error [19] are as follows: cutoff parameter 1000, order 1, and parameter α (not to be confused with the DP's concentration hyperparameter α) set to 2. For the OSPA-E error [20], we use cutoff parameter 120, order 1, and $\alpha = 2$.

BL, SEP, PRO, and KNO use the DAOD algorithm with $\varkappa_{\max} = 2$ for DA and object existence detection.

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